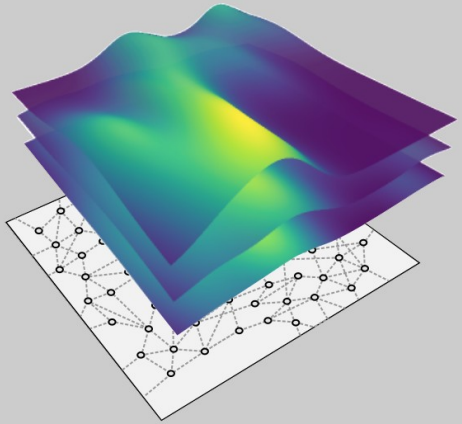
Statistical mapping of structure-function landscapes in microbiomes

Djordje Bajić
Assistant Professor
Department of Biotechnology
TU Delft



School of AI applied to Microbiomes

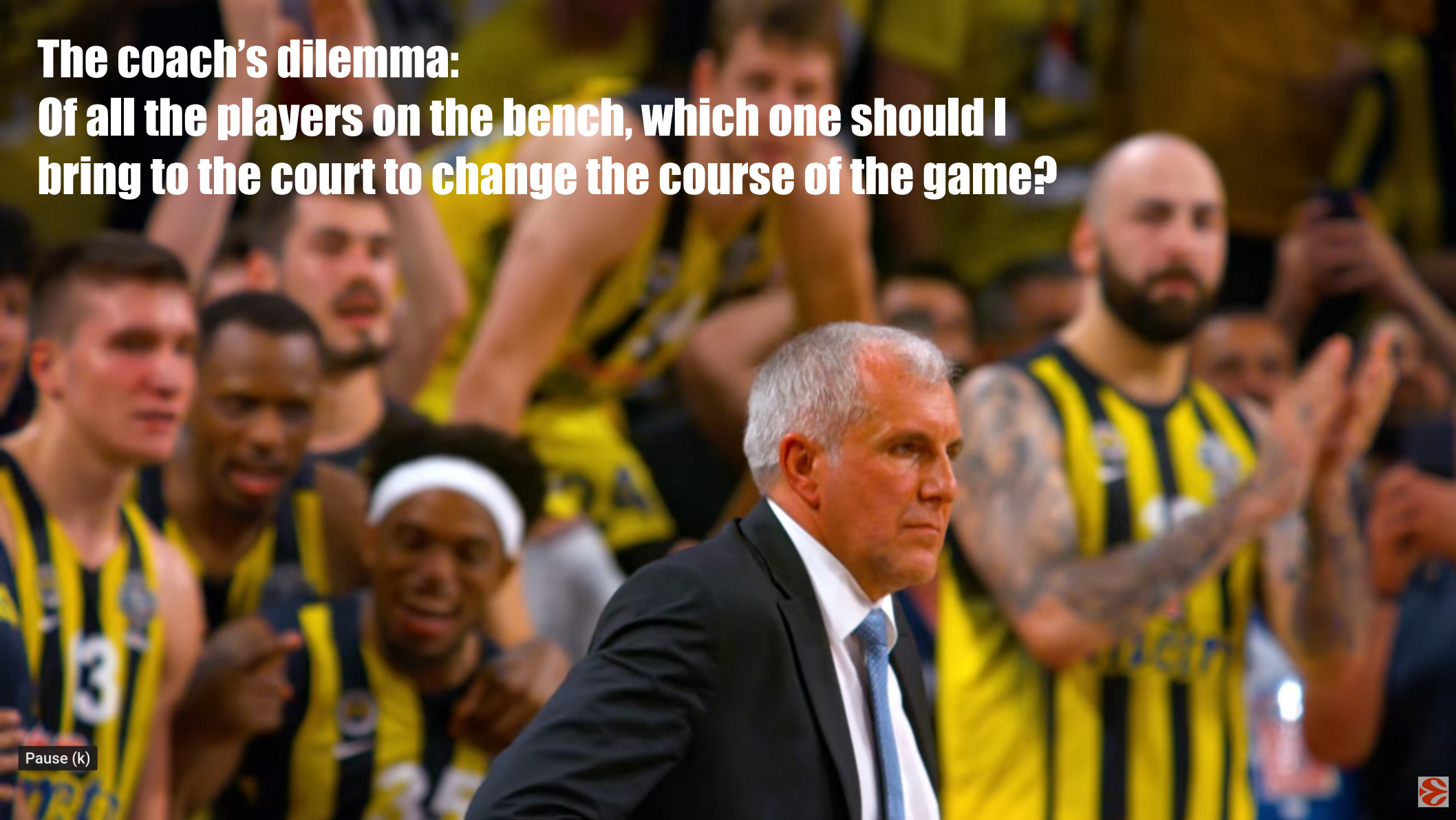
Paris, October 1, 2025

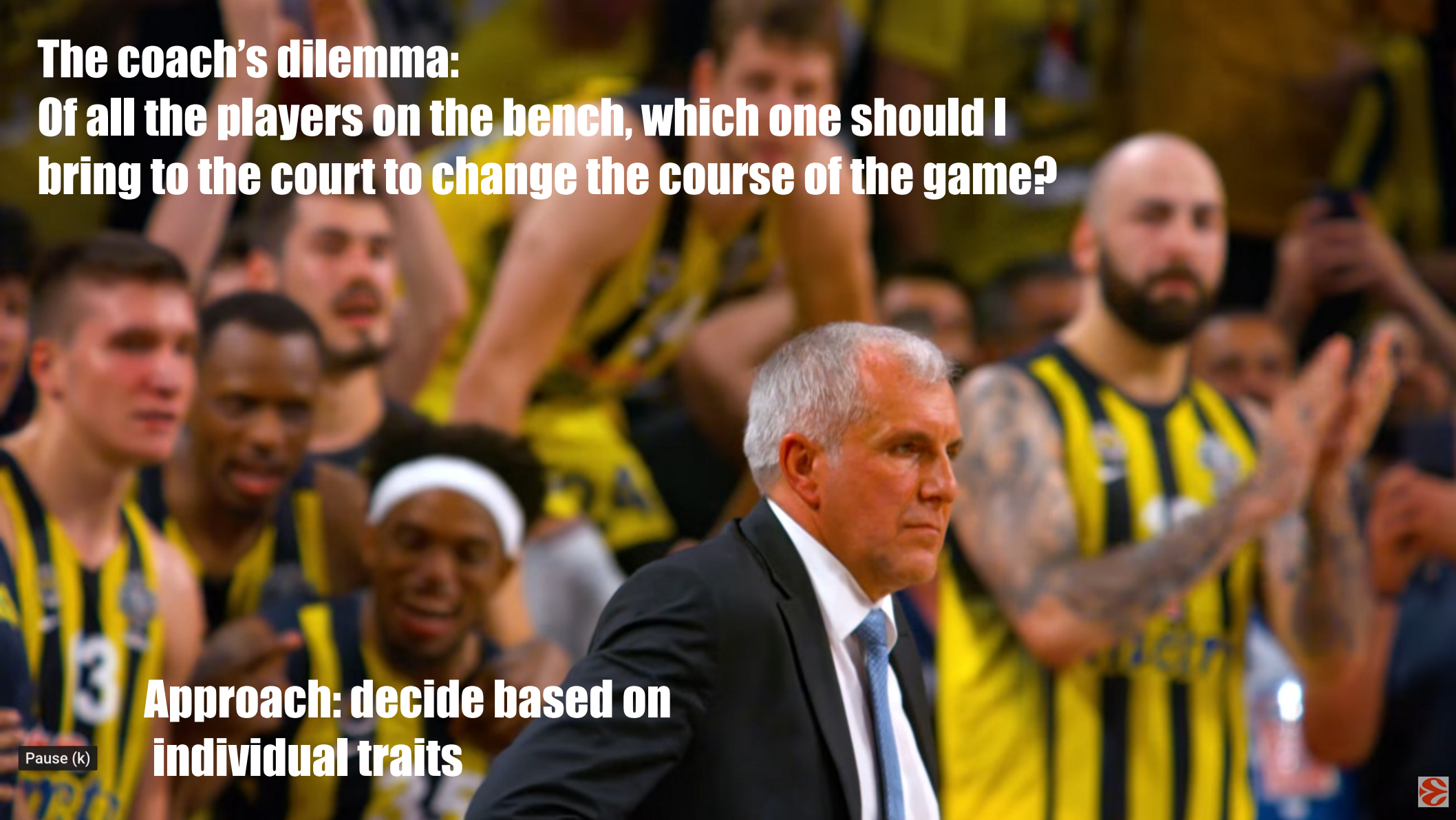
After today session, you will :

1. Know what a structure-function landscape is

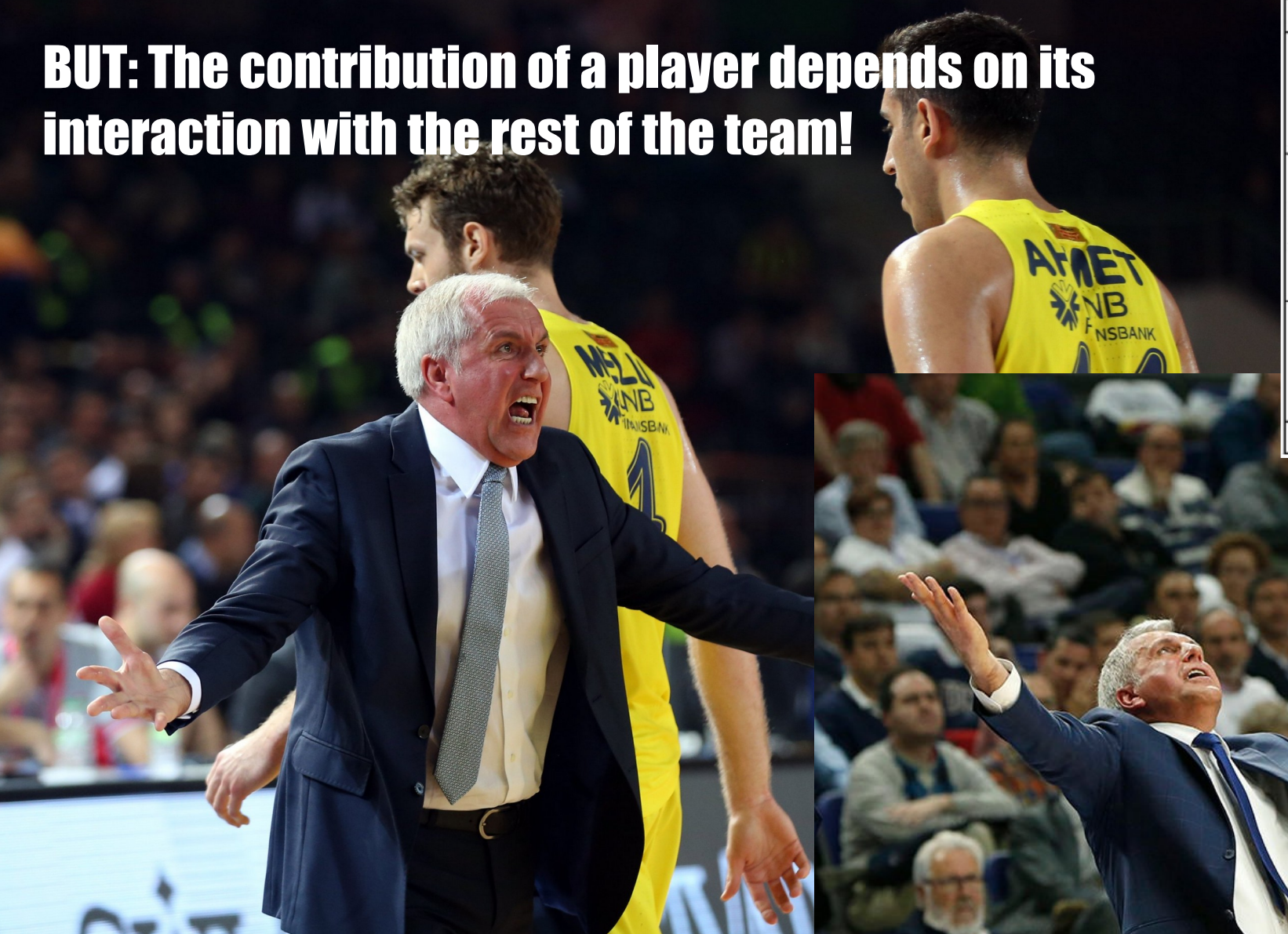
2. Know what global epistasis is and why is useful

3. Touch some machine learning code
   make some predictions
   lose the fear ;)

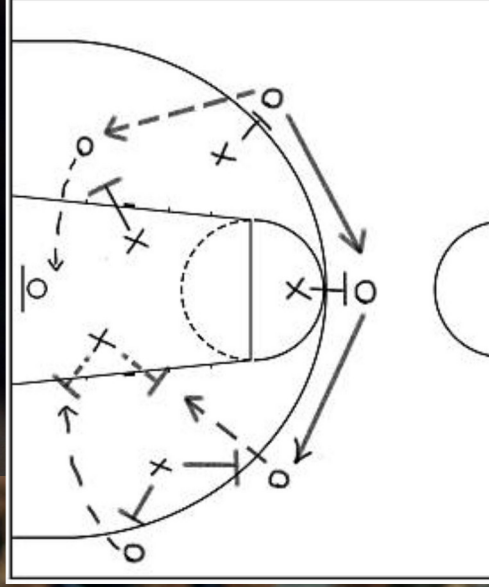

The coach’s dilemma:
Of all the players on the bench, which one should I bring to the court to change the course of the game?
Pause (k)


The coach’s dilemma:
Of all the players on the bench, which one should I bring to the court to change the course of the game?
Approach: decide based on individual traits
Pause (k)

BUT: The contribution of a player depends on its interaction with the rest of the team!



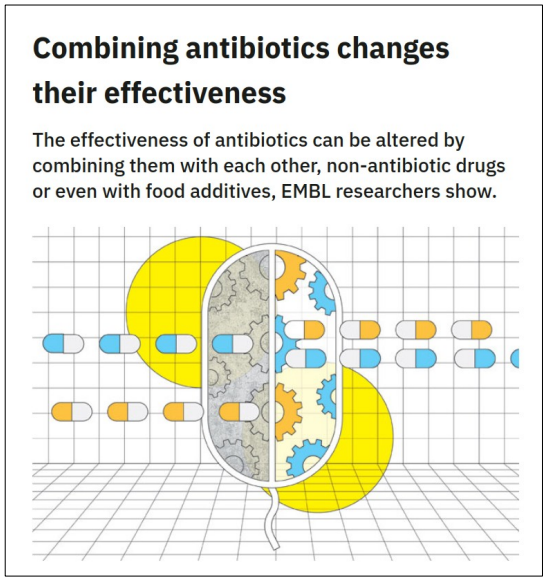




In many systems, function arises from interacting components







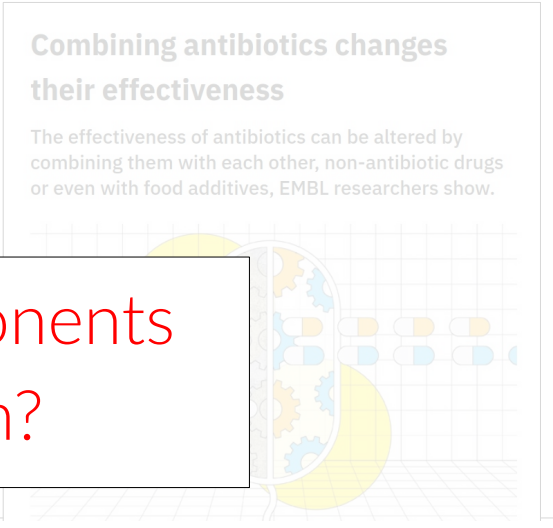


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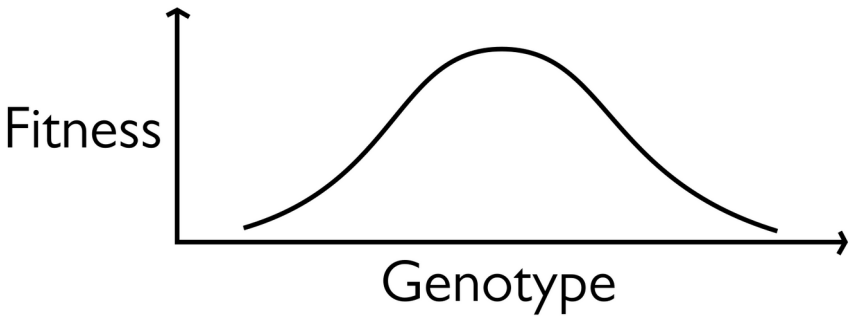
What combination of components maximizes a desired function?



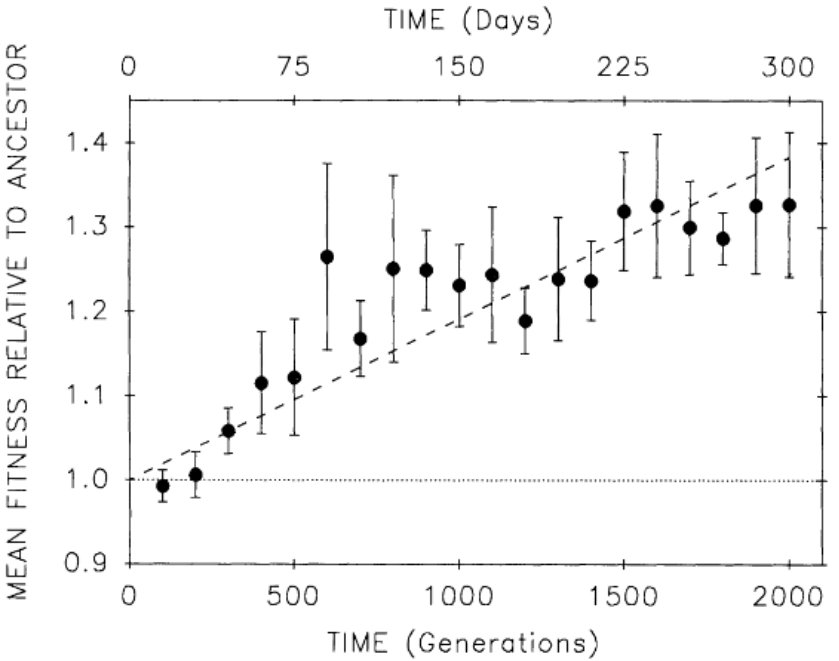
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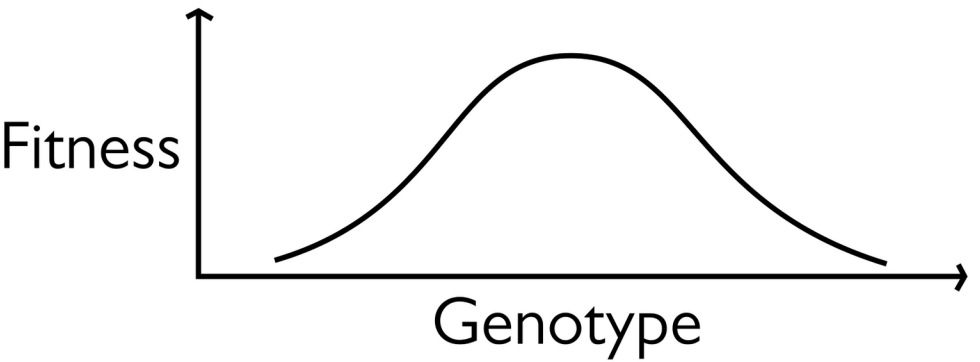
Presenting our main tool:

A **fitness landscape** is a map between genotypes and fitness



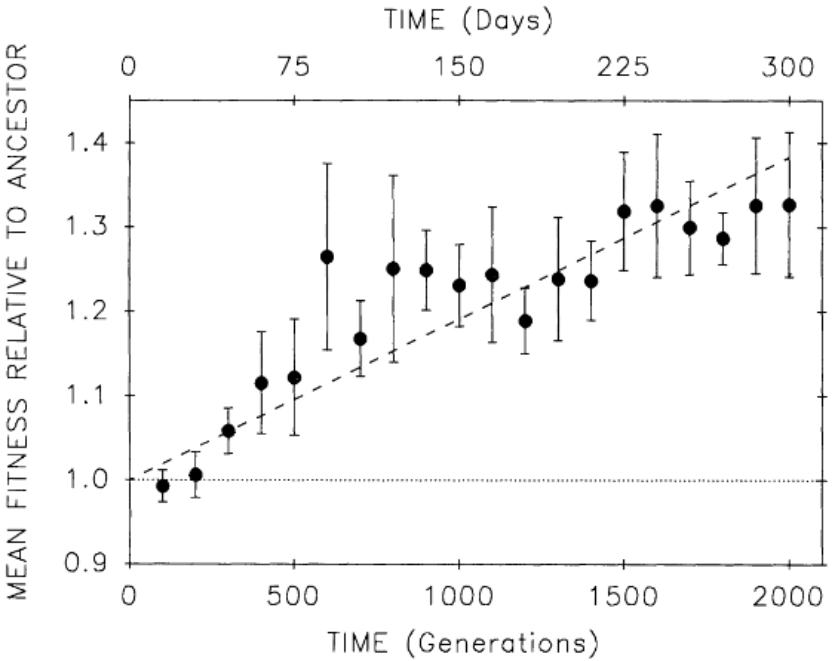
Fitness landscapes have been historically a cornerstone concept to understand and predict evolution

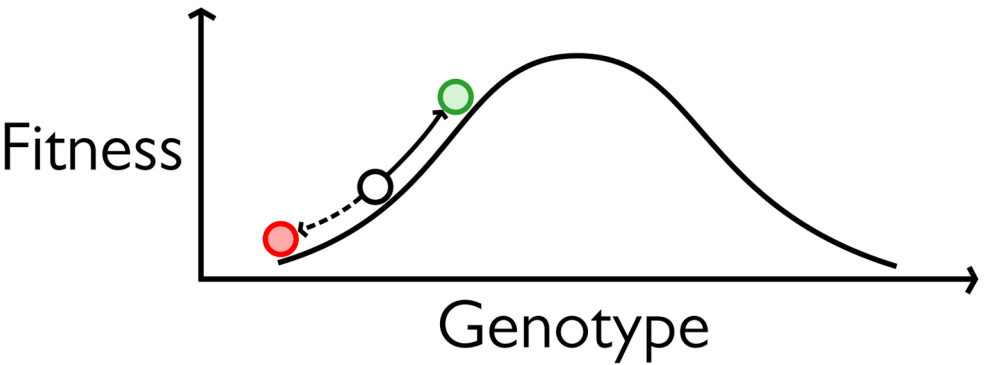




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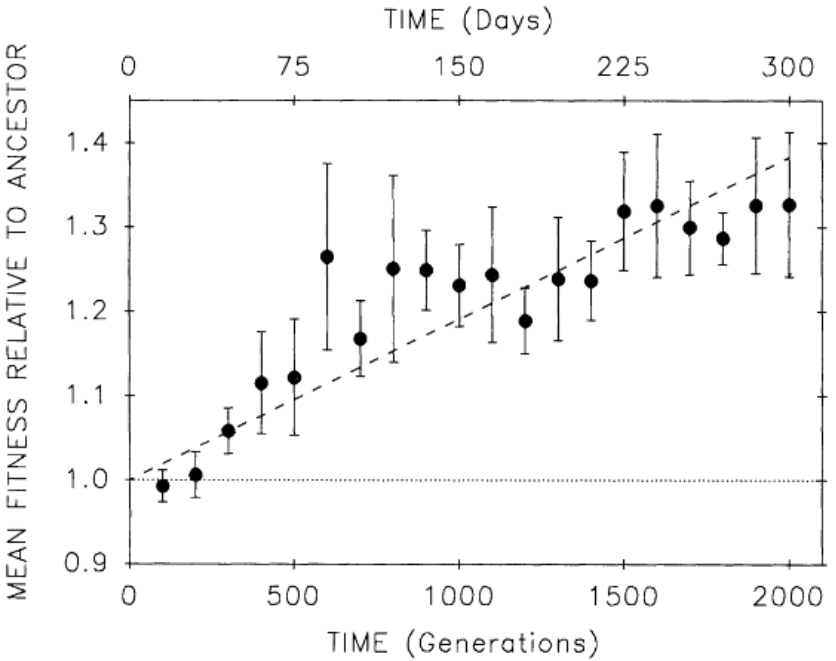
When a spontaneous mutation has higher fitness, it has a chance of fixing. Deleterious mutations get outcompeted

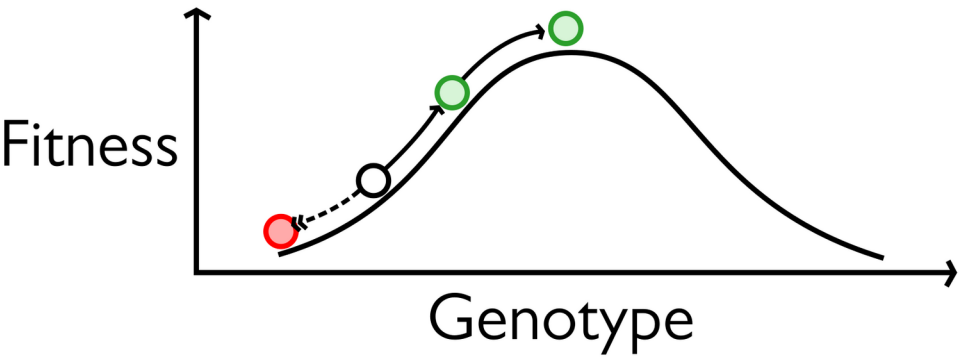




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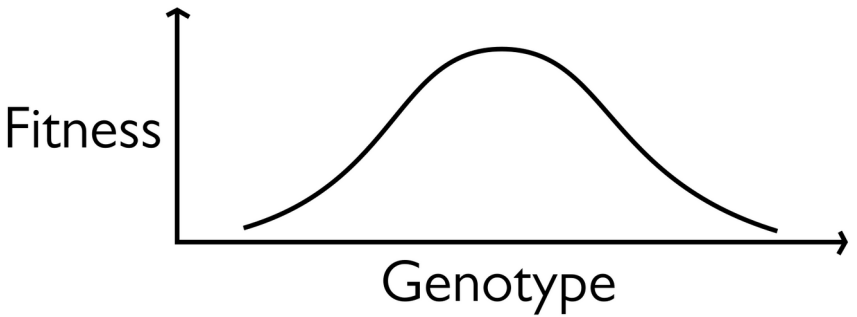
In this way, populations “climb” the peak of the fitness landscape towards genotypes with high fitness

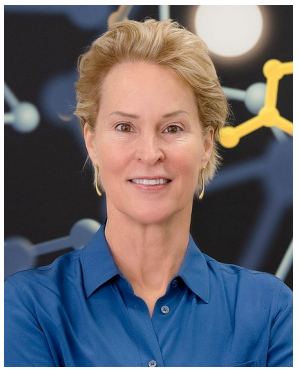




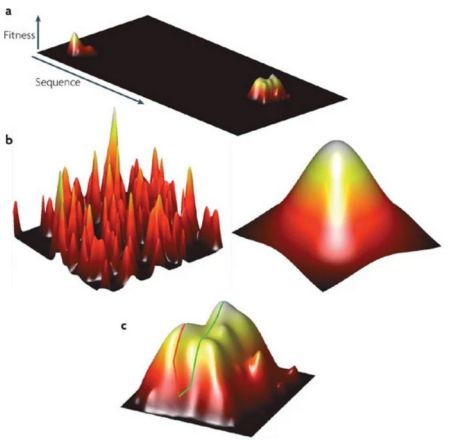
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Fitness landscapes have already been useful for engineering, e.g. proteins



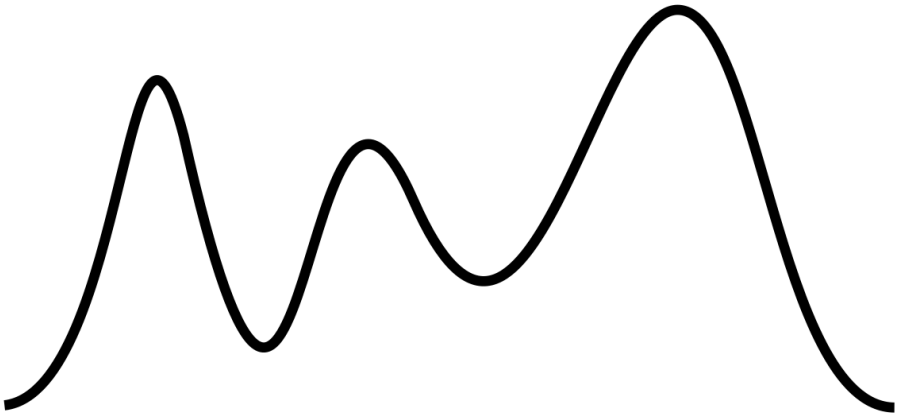


Frances Arnold, Nobel Prize for the directed evolution of enzymes

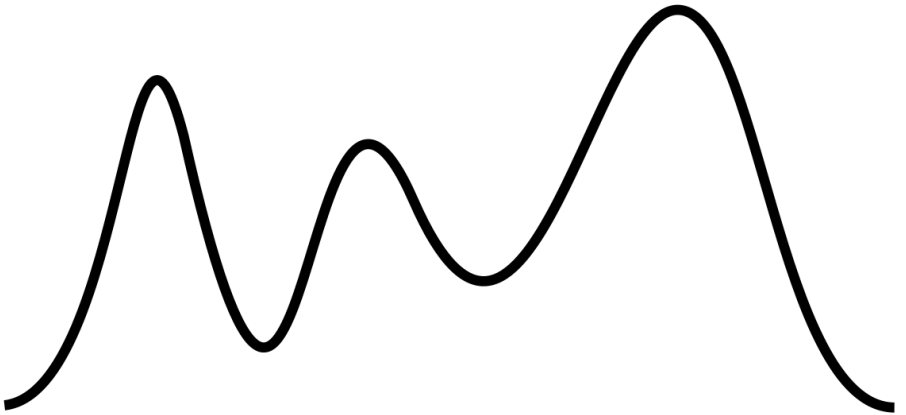


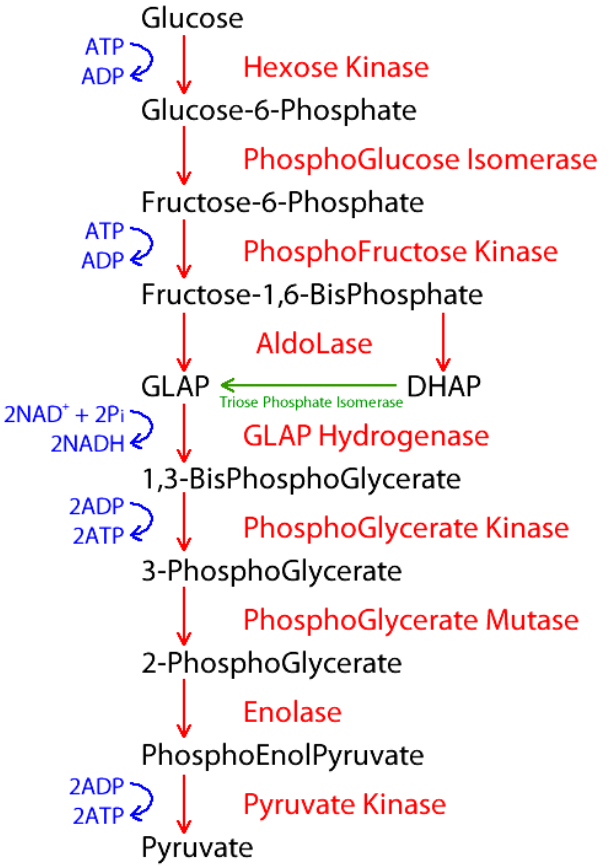
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What determines the shape of a fitness landscape?



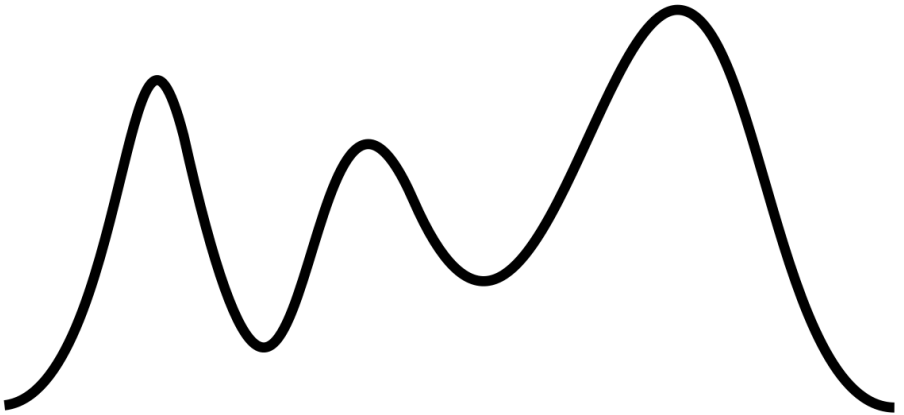
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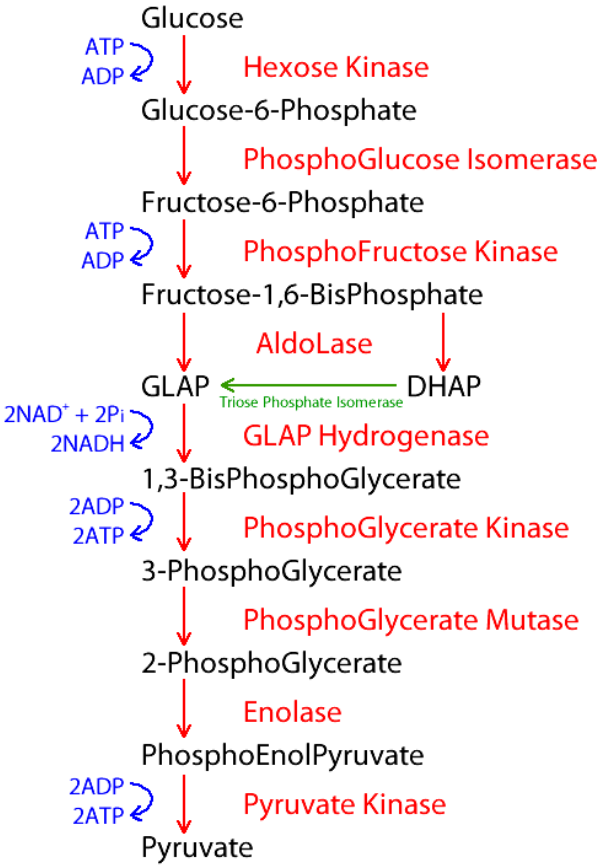


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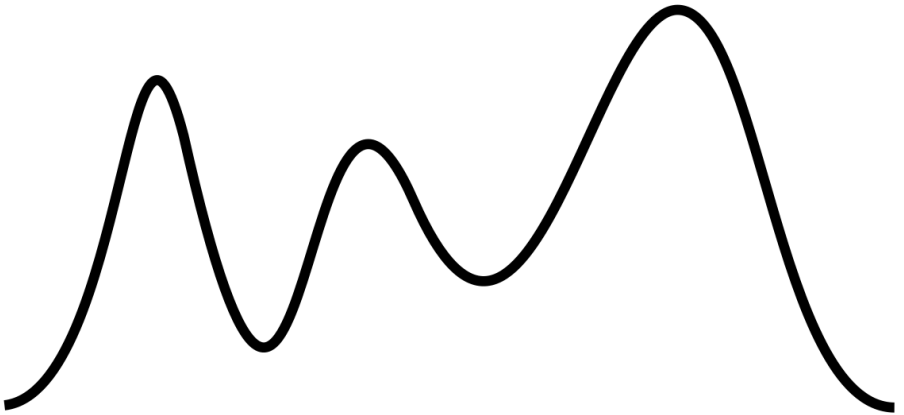


- The environment

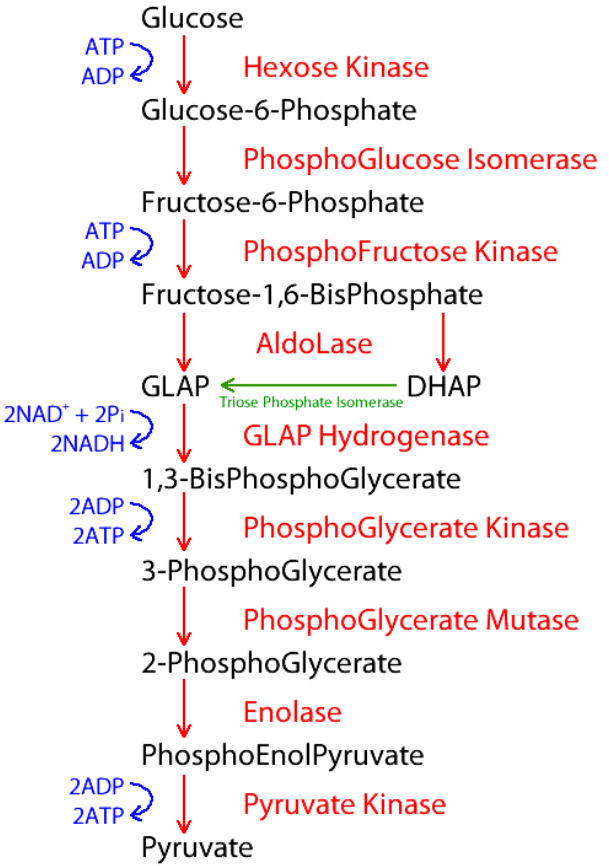


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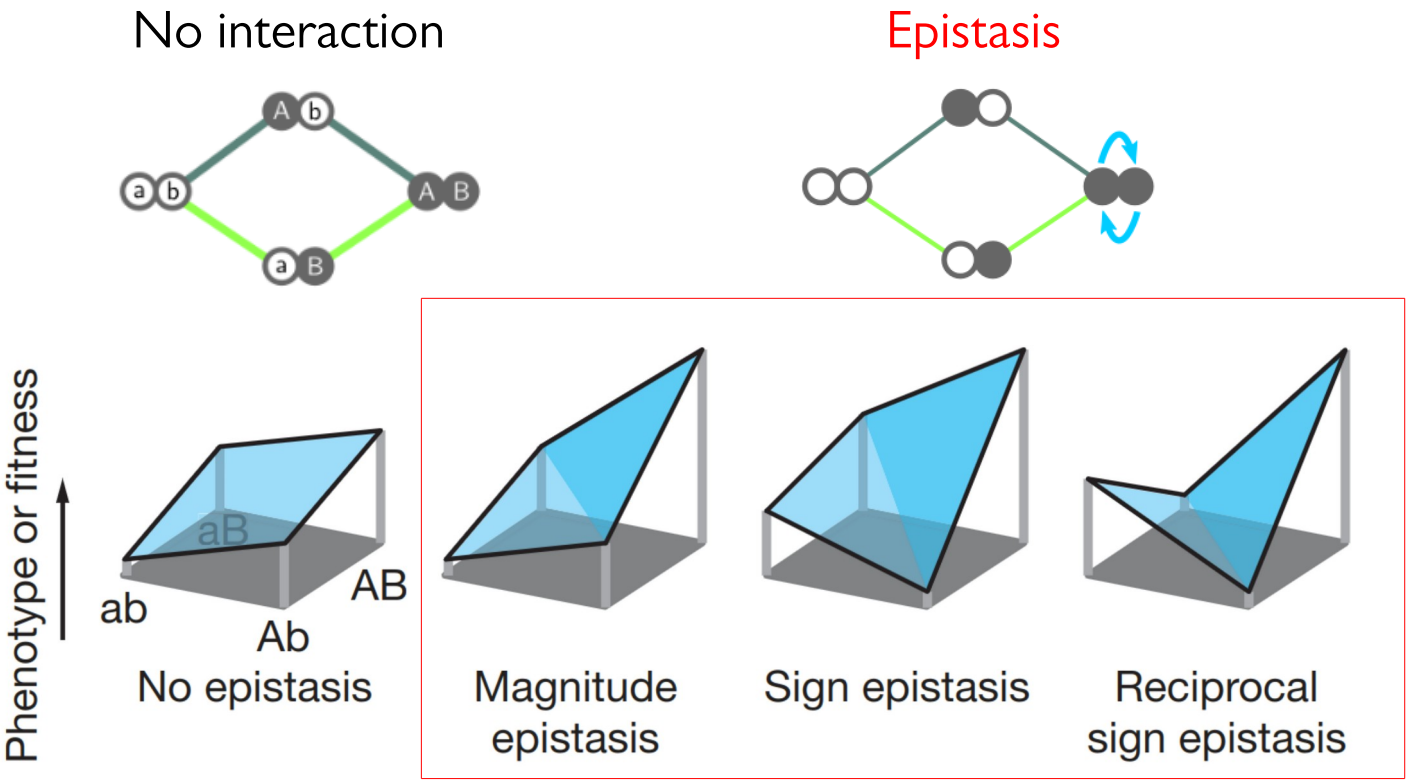


- The environment
- Interactions between mutations - epistasis



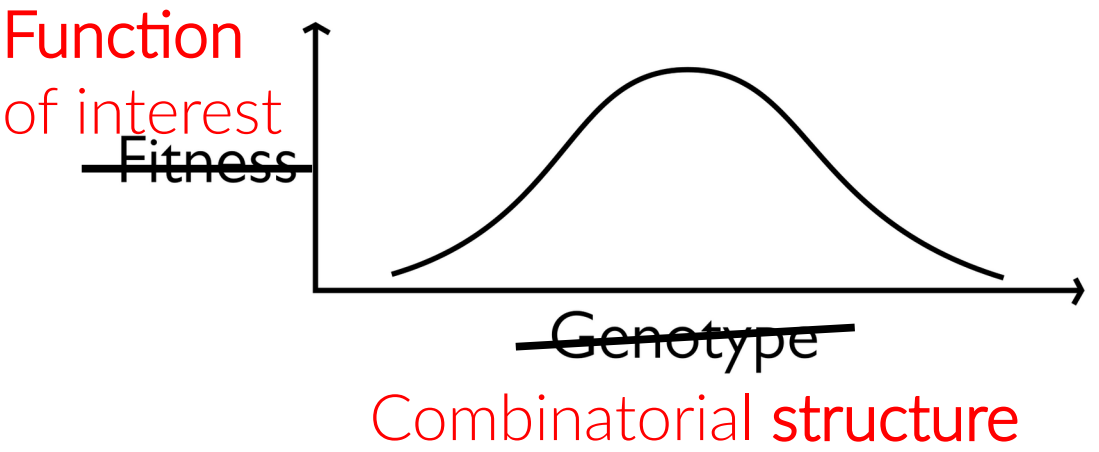
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When one mutation affects the fitness of another one, we say there is an **epistatic interaction** among them

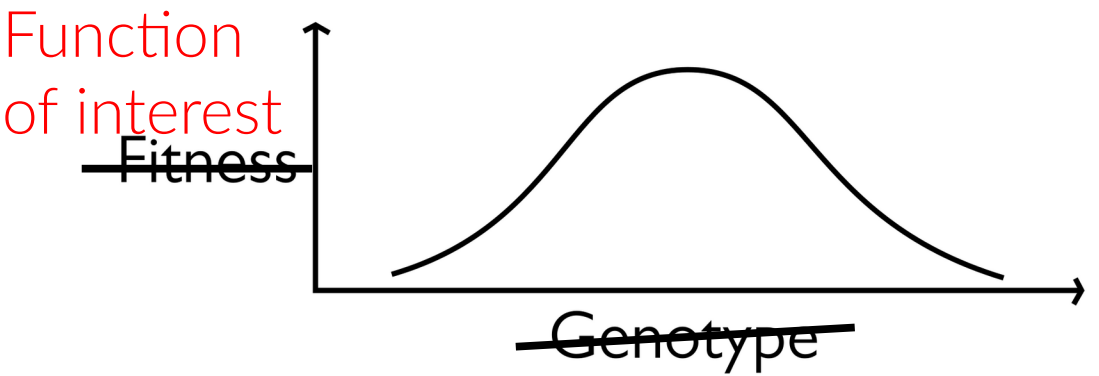


Poelwijk et al, 2007 "**Empirical fitness landscapes reveal accessible evolutionary paths**", Nature

Can we extend the theory of fitness landscapes to engineer other systems with interacting components?



Can we extend the theory of fitness landscapes to engineer other systems with interacting components?

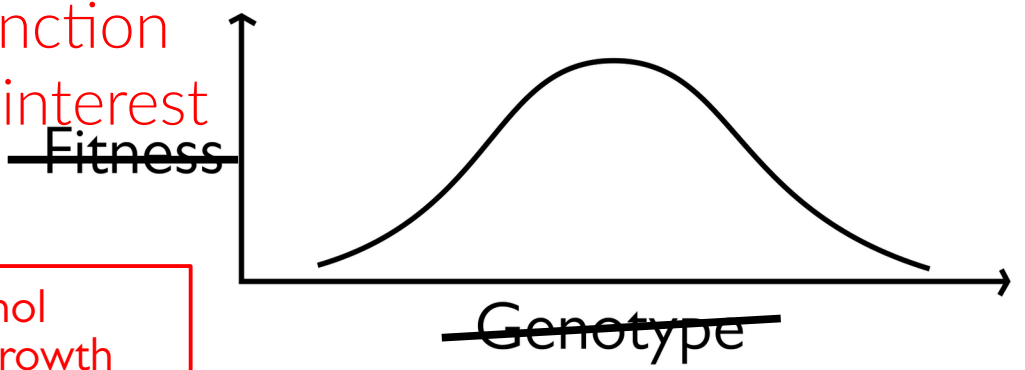


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Can we extend the theory of fitness landscapes to engineer other systems with interacting components?

Function of interest

- production of ethanol
- effect on a plant's growth
- whatever you can quantify

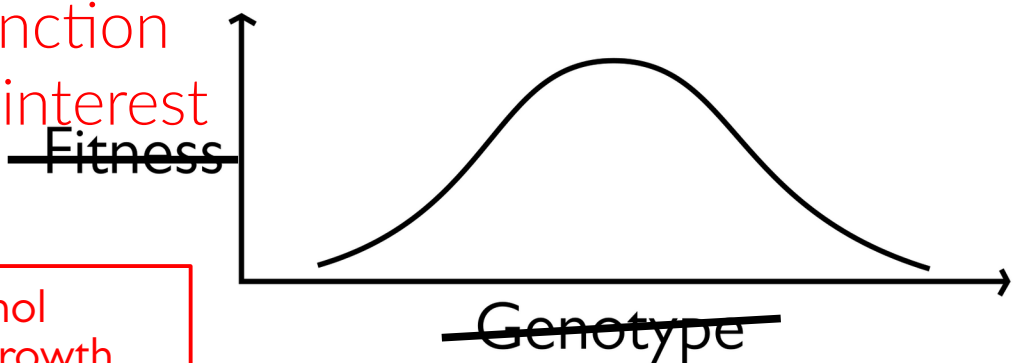


Combinatorial structure

The composition of a synthetic microbial community

A structure-function landscape

Function of interest

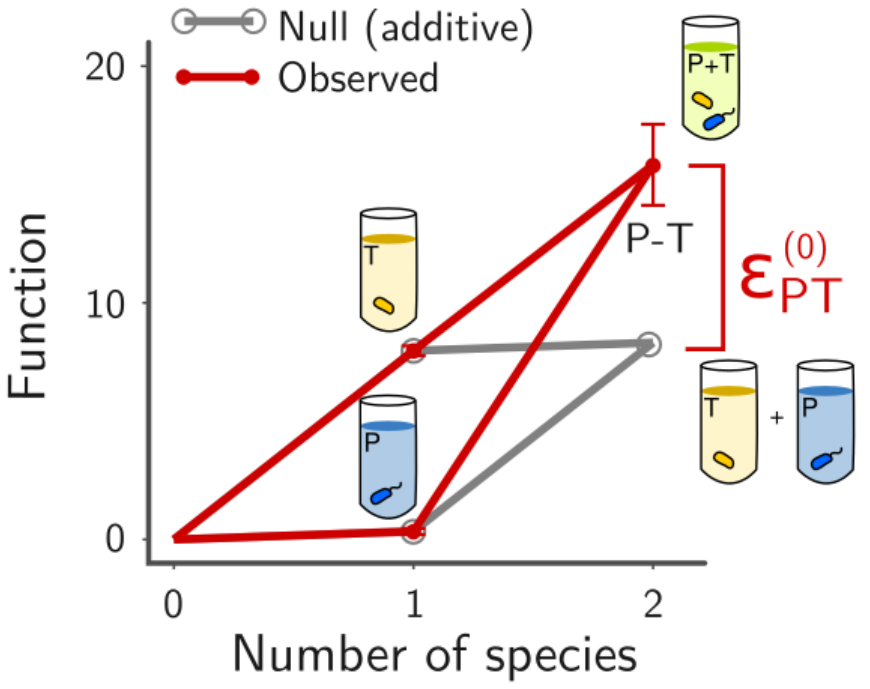


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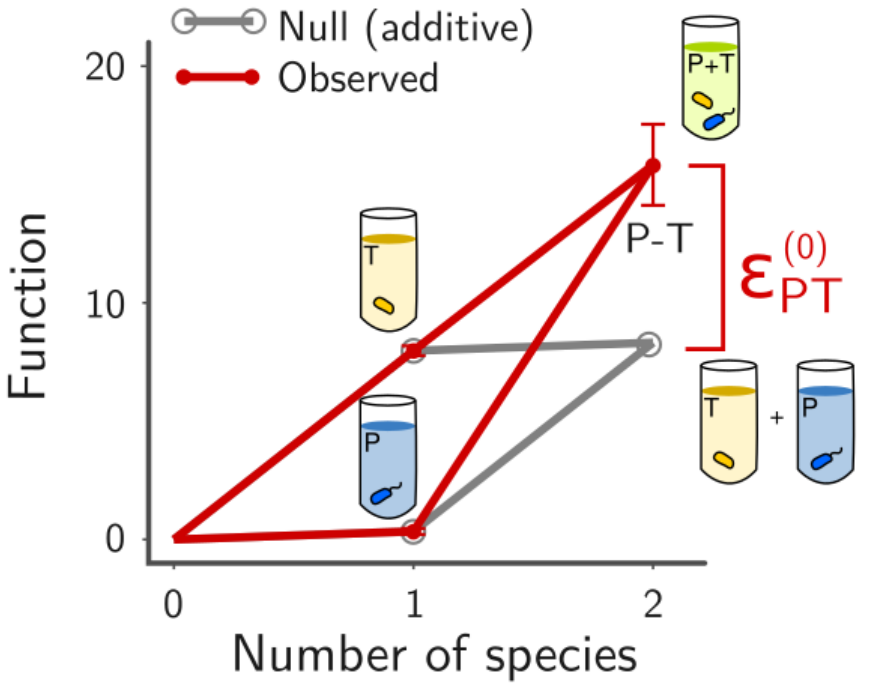
Combinatorial structure

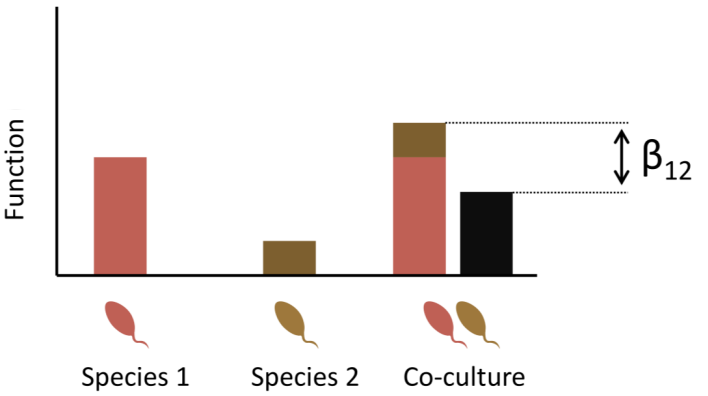
Example: the composition of a synthetic microbial community

Same as fitness landscapes, structure-function landscapes are also shaped by **interactions**



Same as fitness landscapes, structure-function landscapes are also shaped by **interactions**



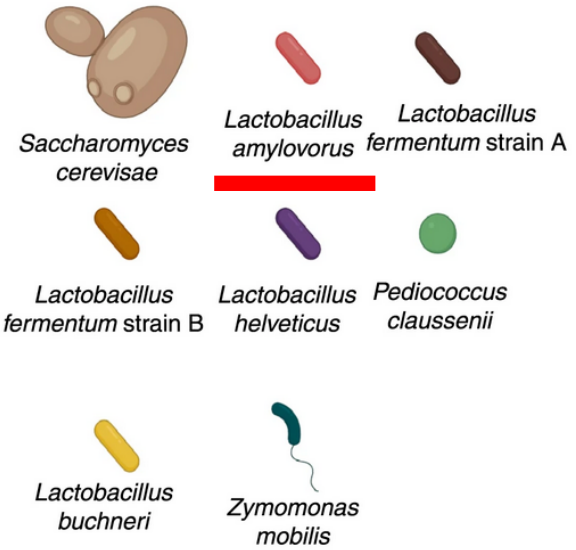


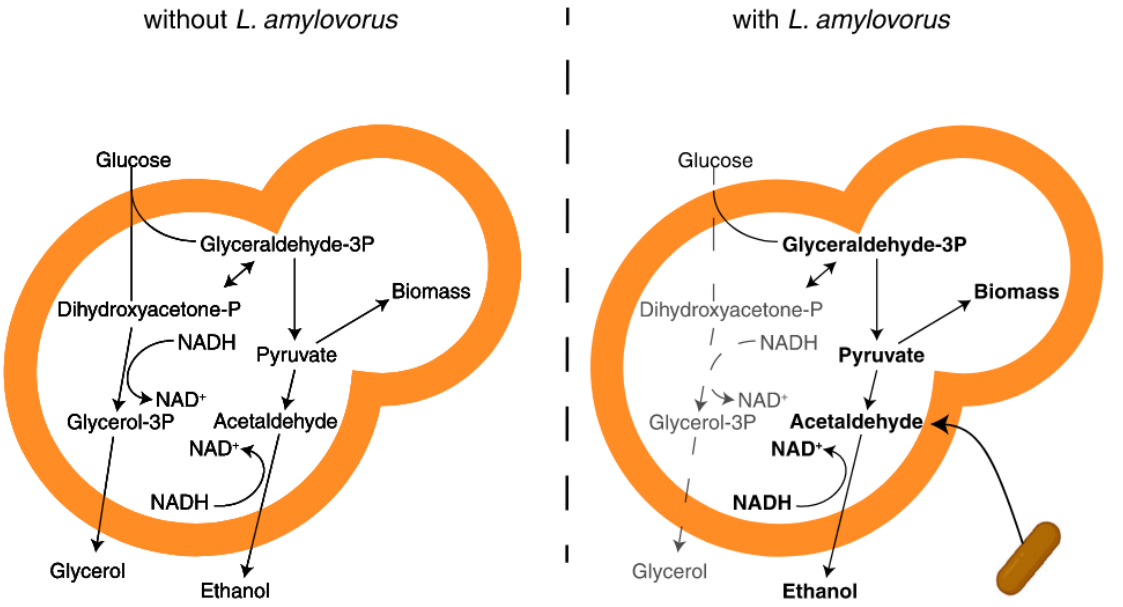
$$F(x_1, x_2) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2$$

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Structure-function landscape modeling helped identify a potential industrial probiotic







nature COMMUNICATIONS

Complex yeast–bacteria interactions affect the yield of industrial ethanol fermentation

Felipe Senne de Oliveira Lino[1], Djordje Bajic[2,3], Jean Celestin Charles Vila[2,3], Alvaro Sánchez[2,3] & Morten Otto Alexander Sommer[1✉]



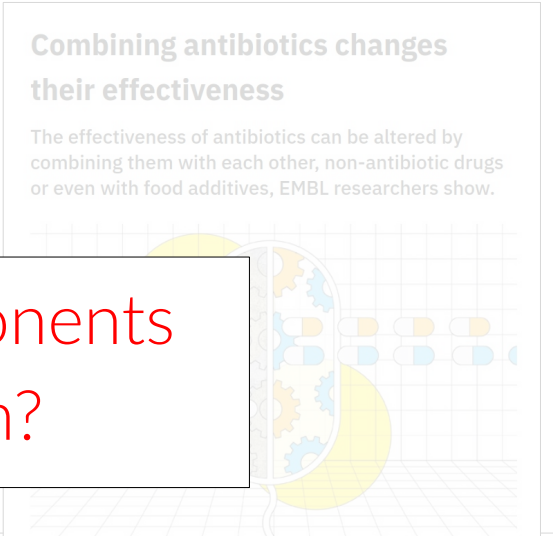
Felipe S. Lino

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In systems where function arises from interacting components…

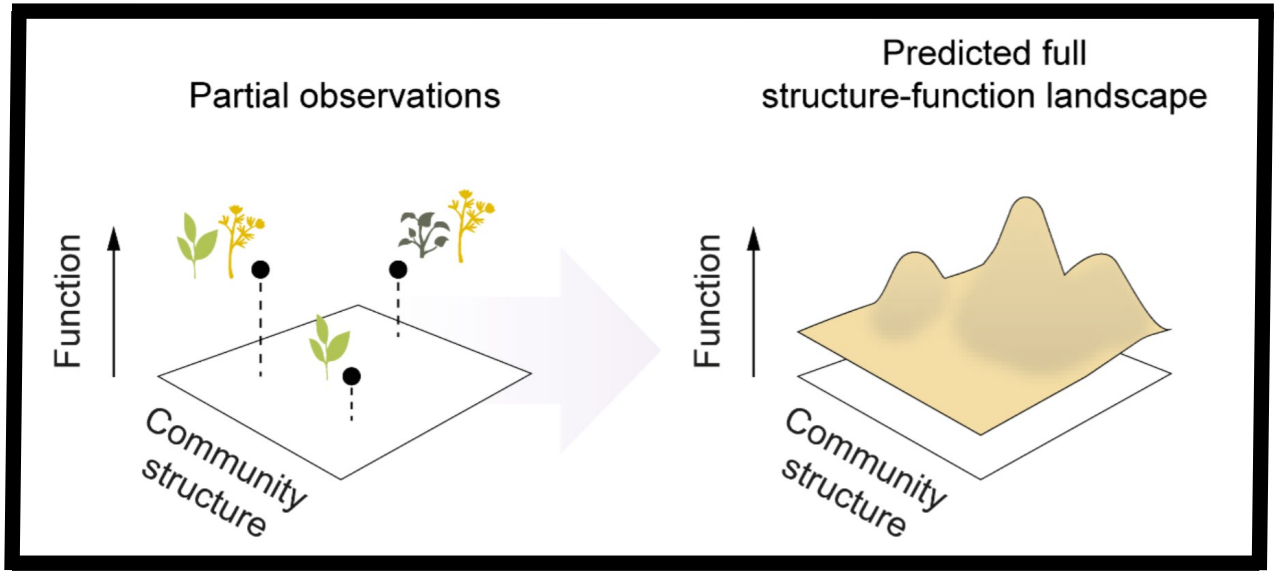


What combination of components maximizes a desired function?



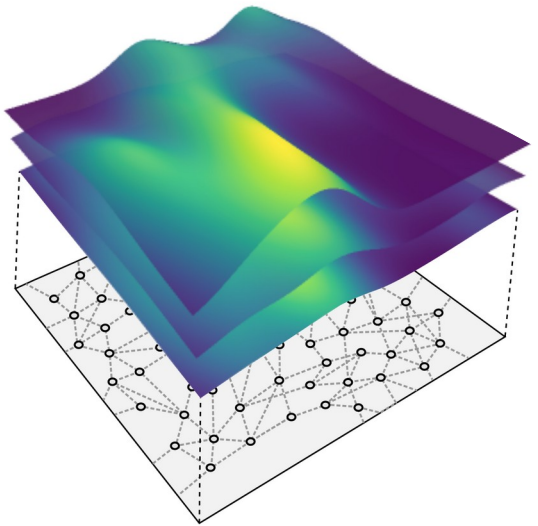
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Our goal: predict the structure-function landscape of a microbial community **with limited data, because the combinatorial space is infeasible!** (2^n)



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Simplest statistical approach: regression



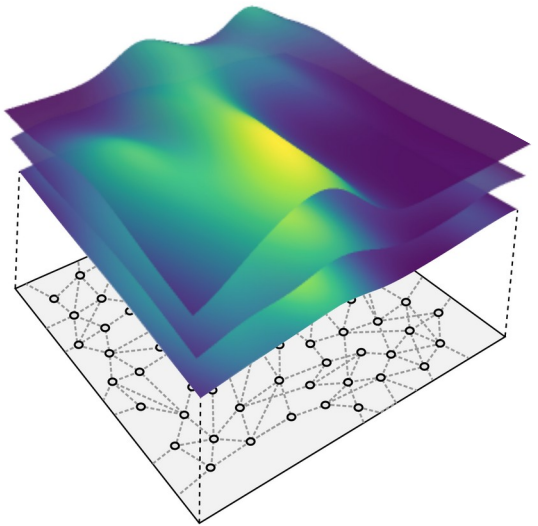
$$y = \beta_0 + \sum_i \beta_i x_i$$

Community function

Effect of species i on function

Presence/absence of species i in the community

Simplest statistical approach: regression



$$y = \beta_0 + \sum_i \beta_i x_i + \sum_{i<j} \beta_{ij} x_i x_j + \ldots,$$

Community function

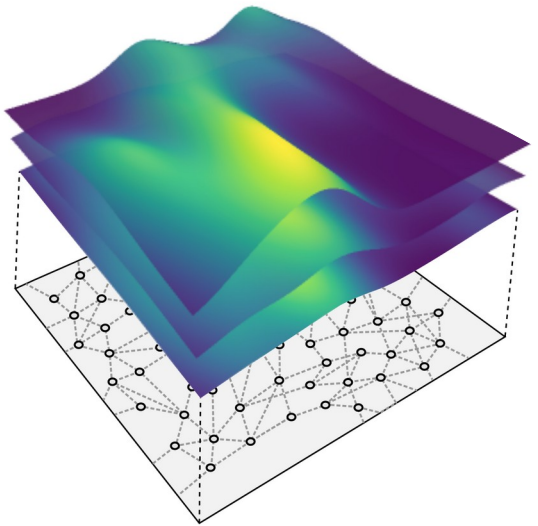
Effect of species i on function

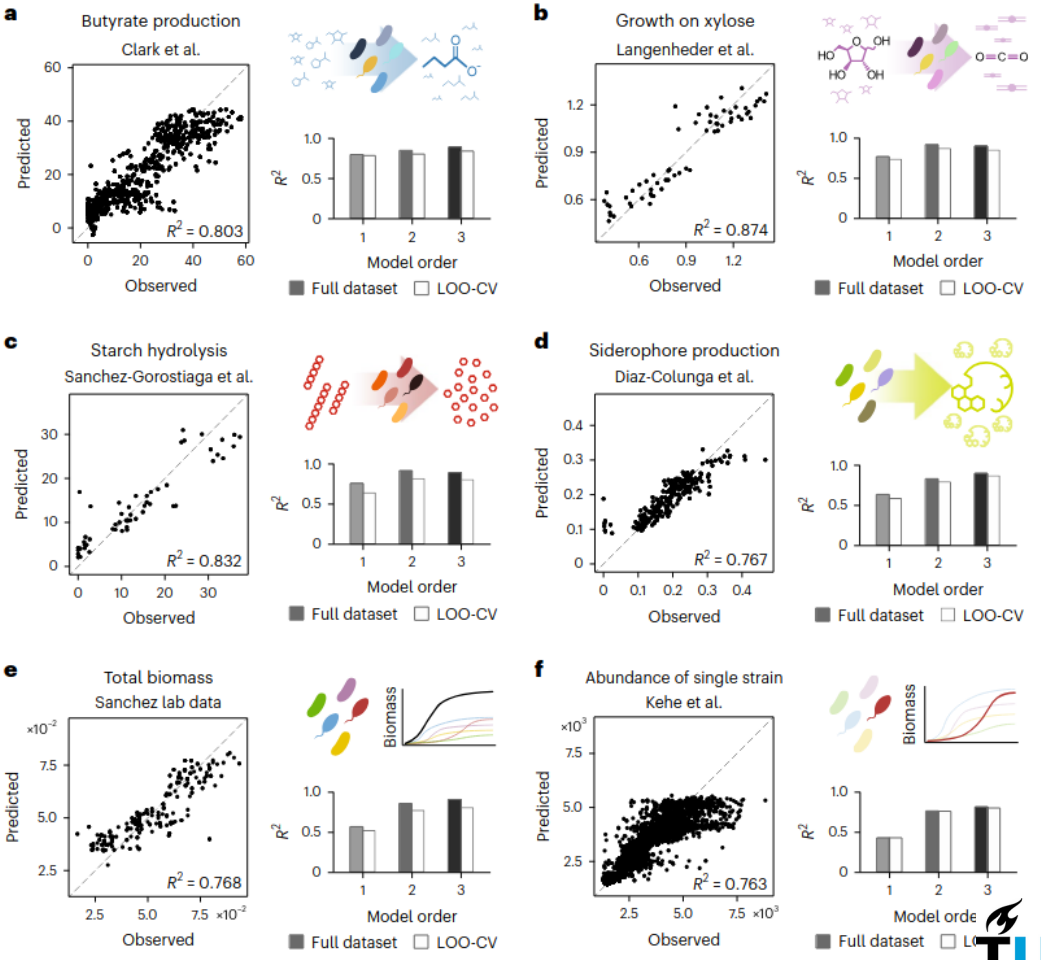
Presence/absence of species i in the community

Effect of species i and j When both present (**interaction**)

Linear regression works quite OK!

$$y = \beta_0 + \sum_i \beta_i x_i + \sum_{i<j} \beta_{ij} x_i x_j + \dots,$$

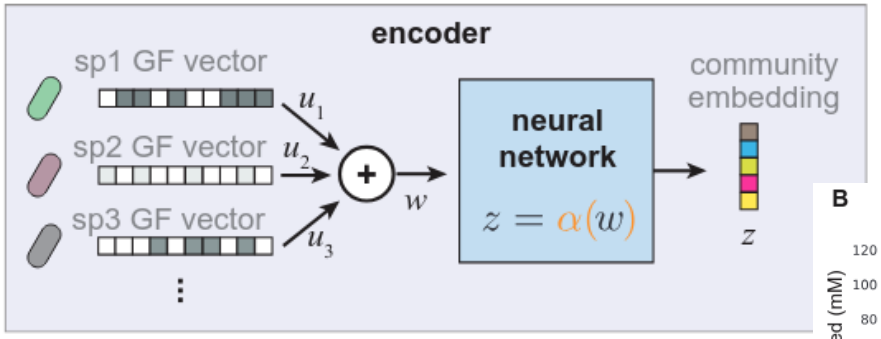


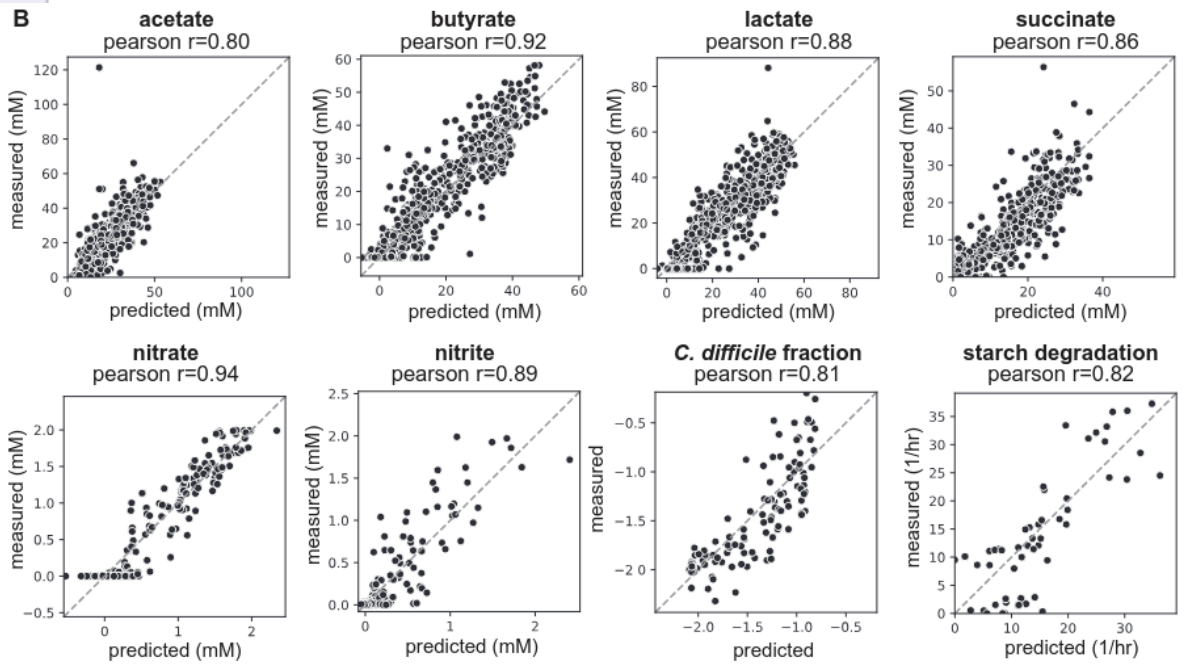


Skwara et al, Nat. Eco. Evo. 2023

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More complex approaches: machine learning





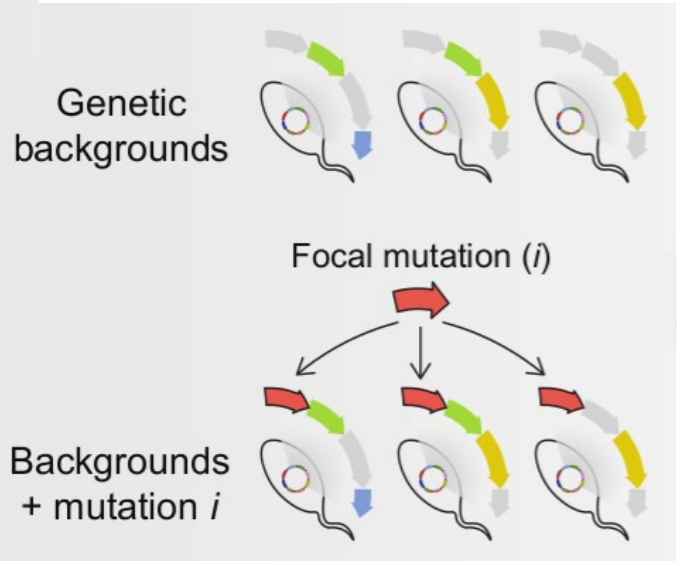
Qian et al, biorxiv 2025

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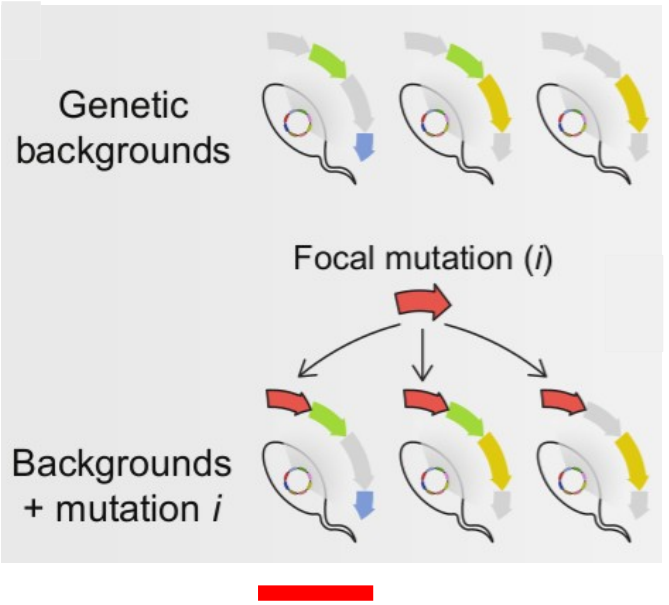
Global epistasis:
A new way to look at interactions

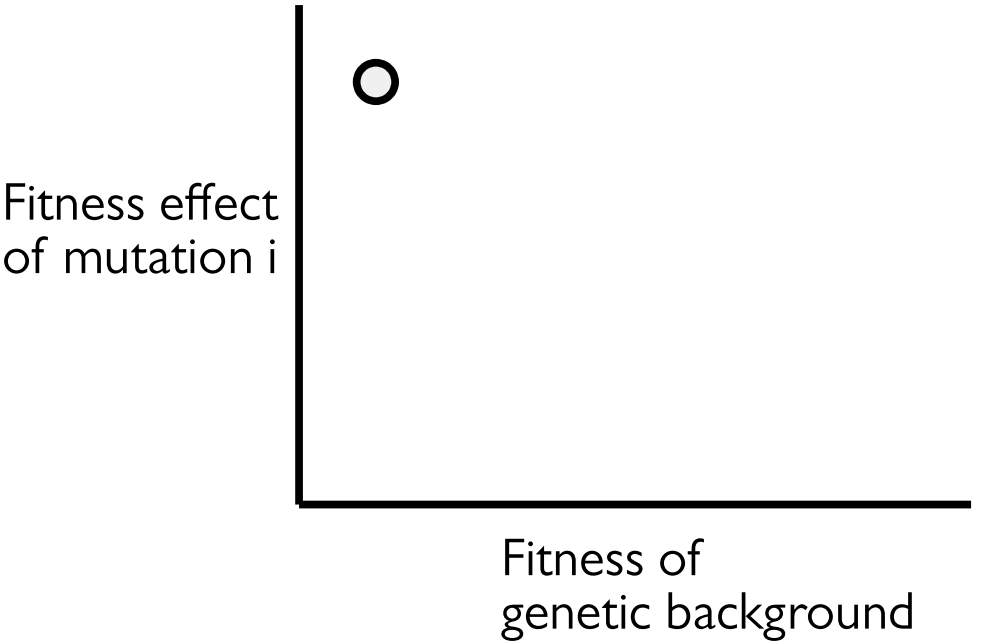
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Back to genomes: let's test the effect of a mutation, one background at a time

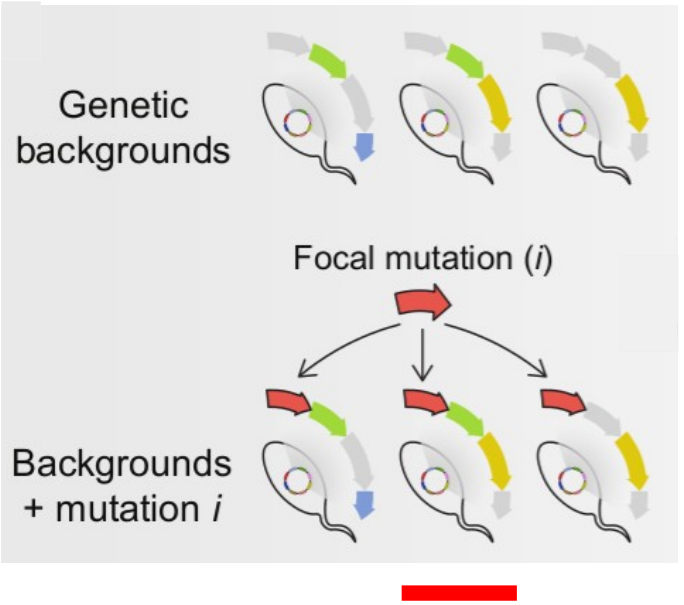


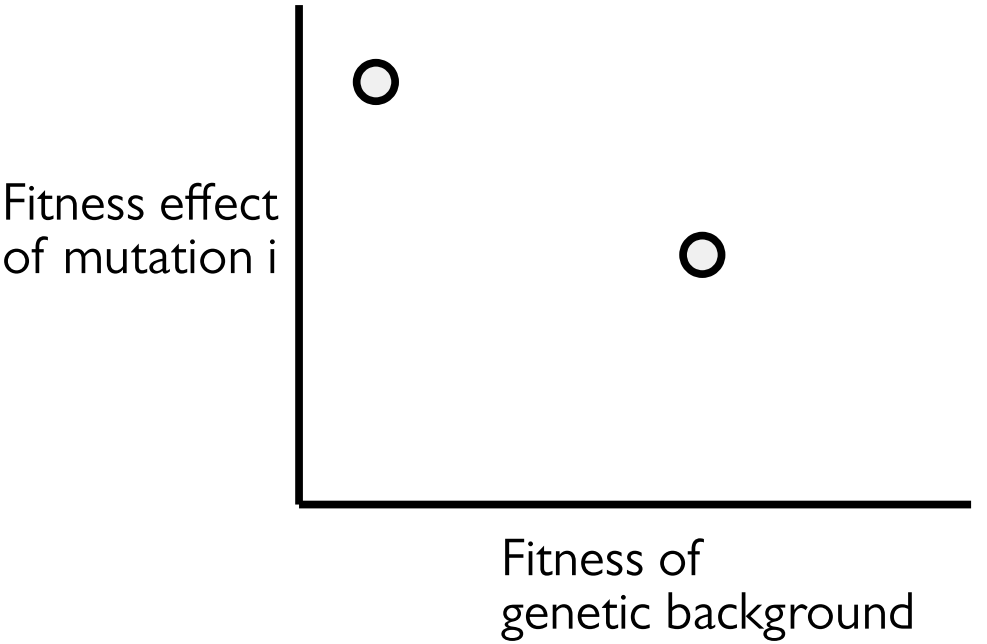
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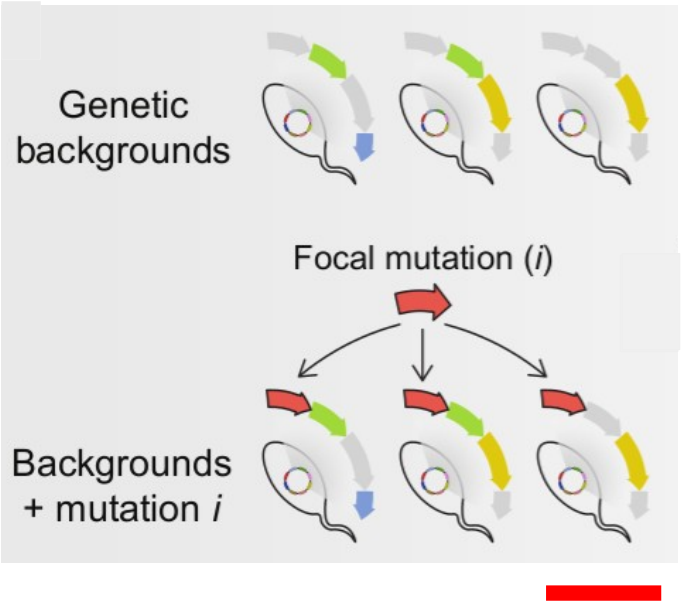


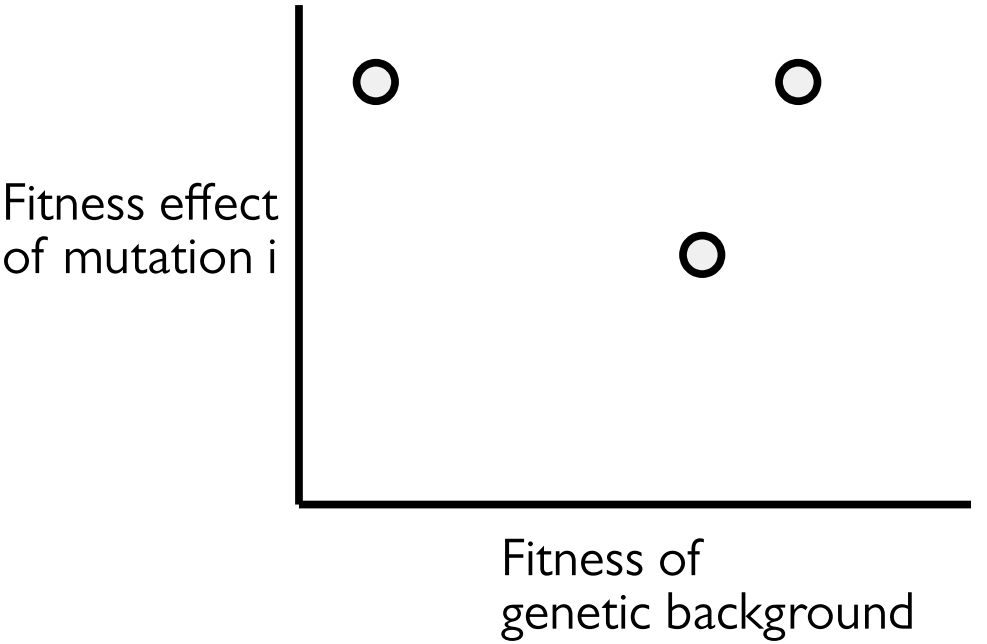
Testing a mutation, one background at a time



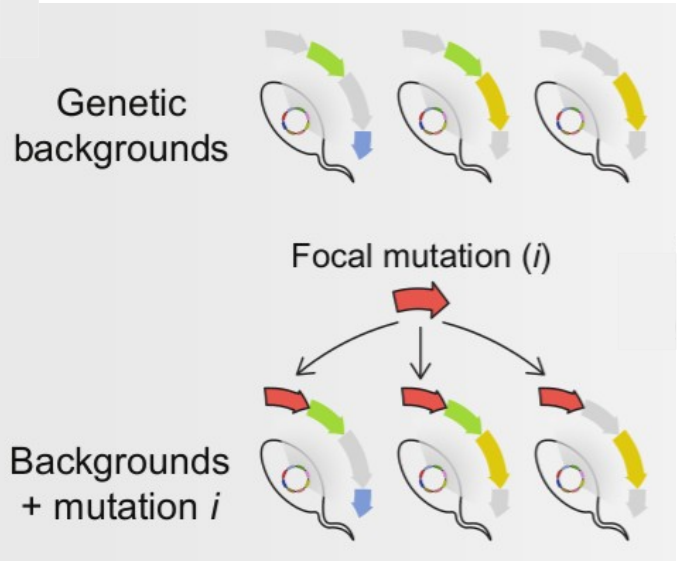


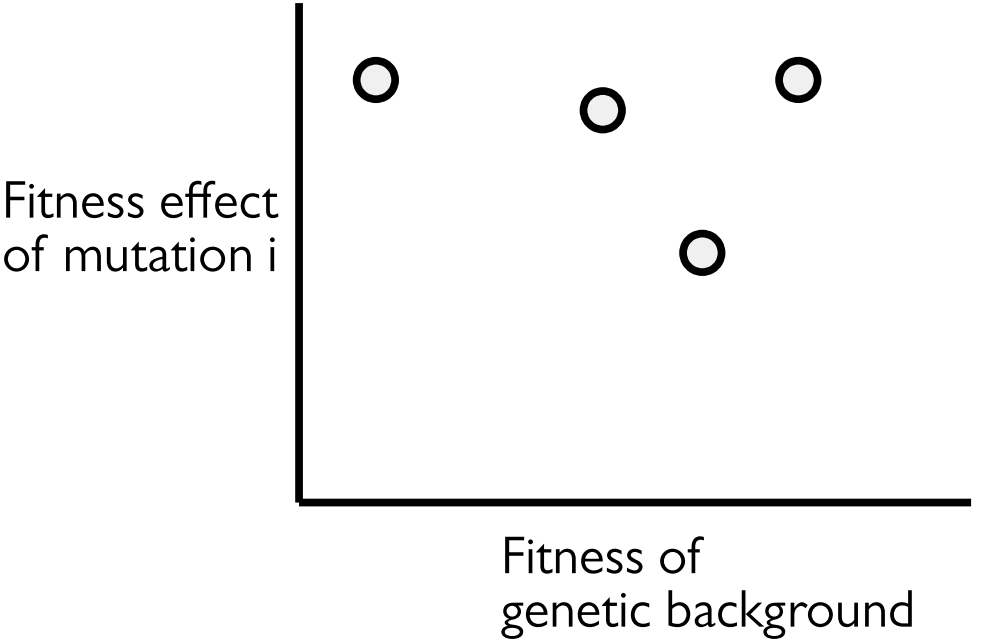
Testing a mutation, one background at a time



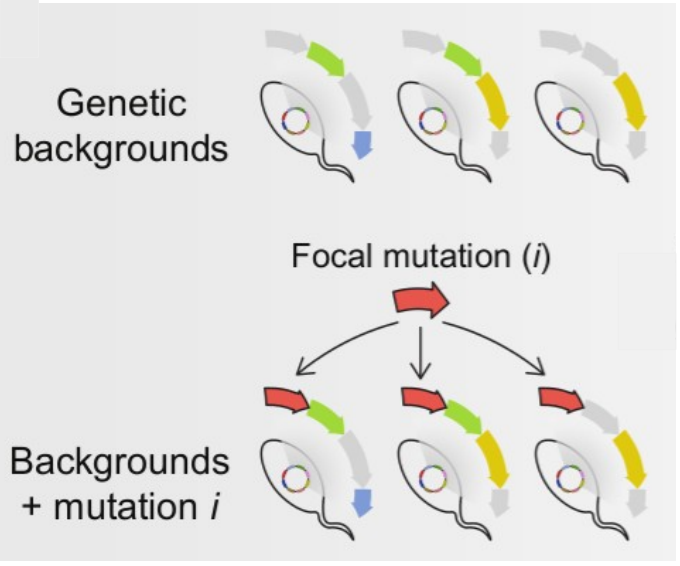


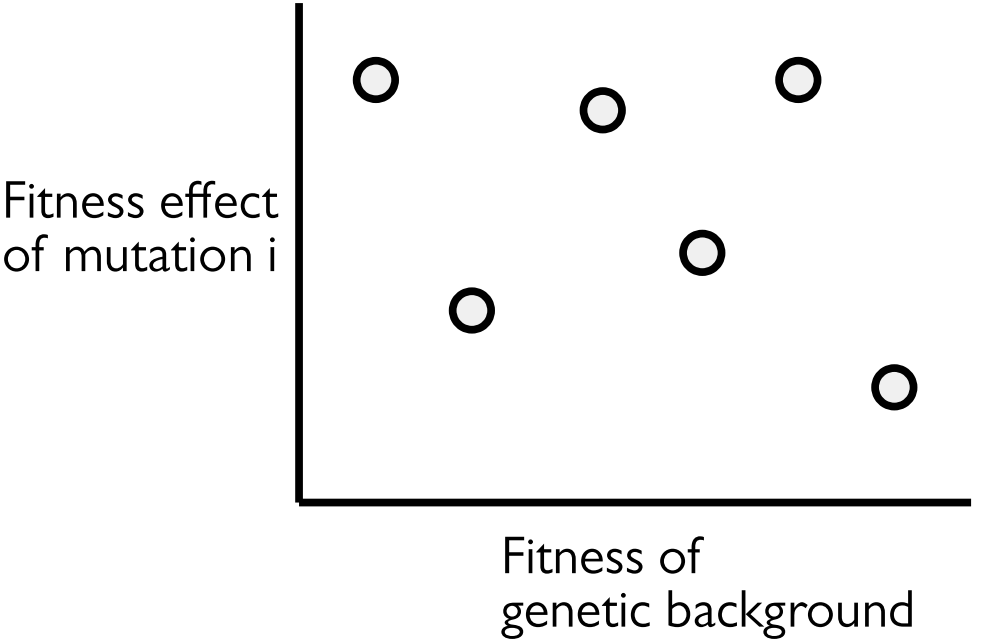
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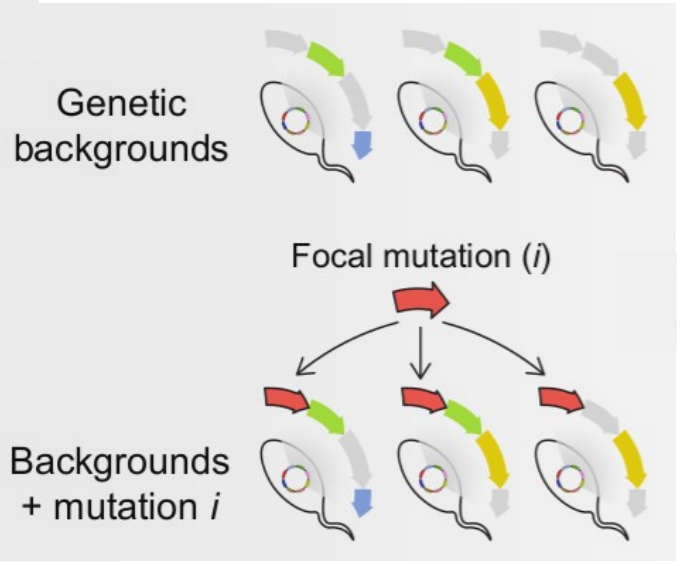


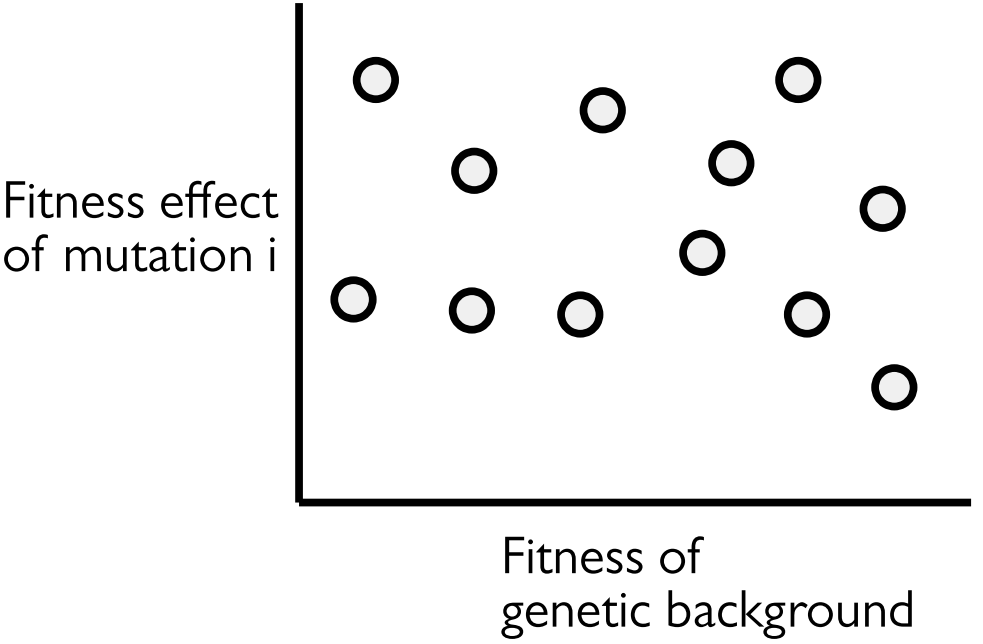
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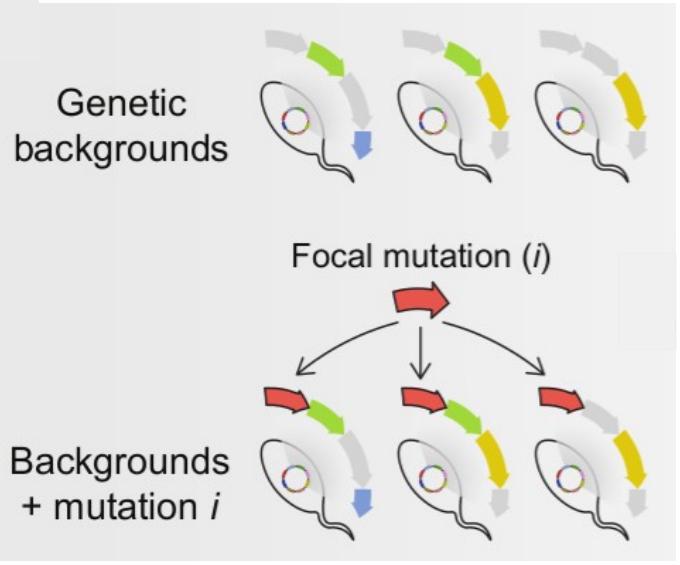


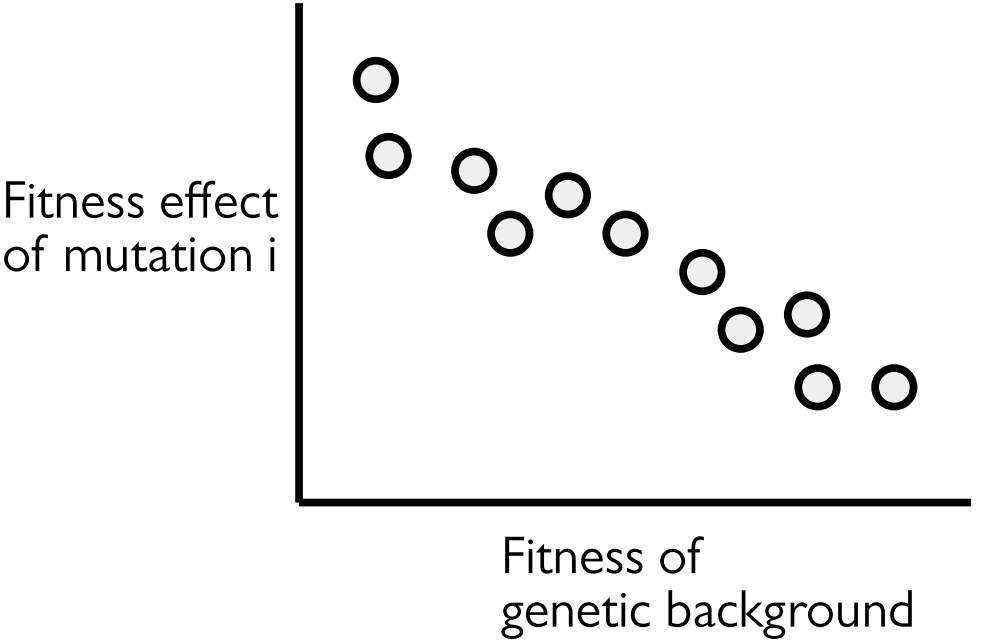
Intuitively, interactions will lead to rugged landscapes, where making predictions is hard





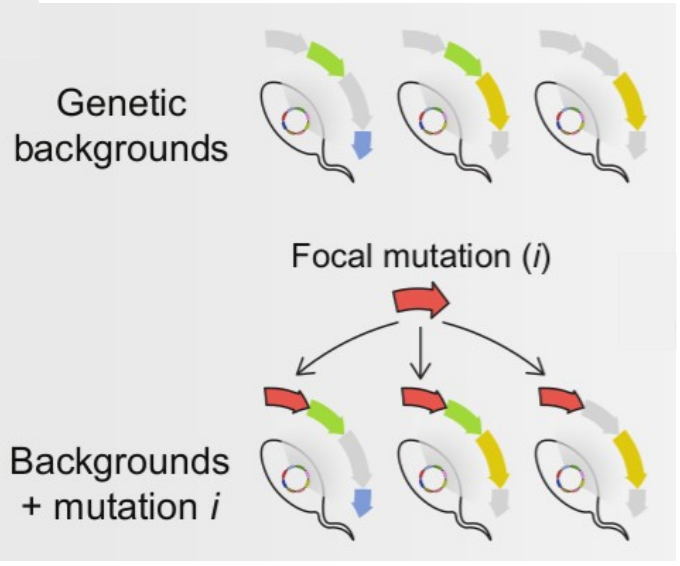
Global epistasis is an emergent linear scaling pattern found in many empirical fitness landscapes

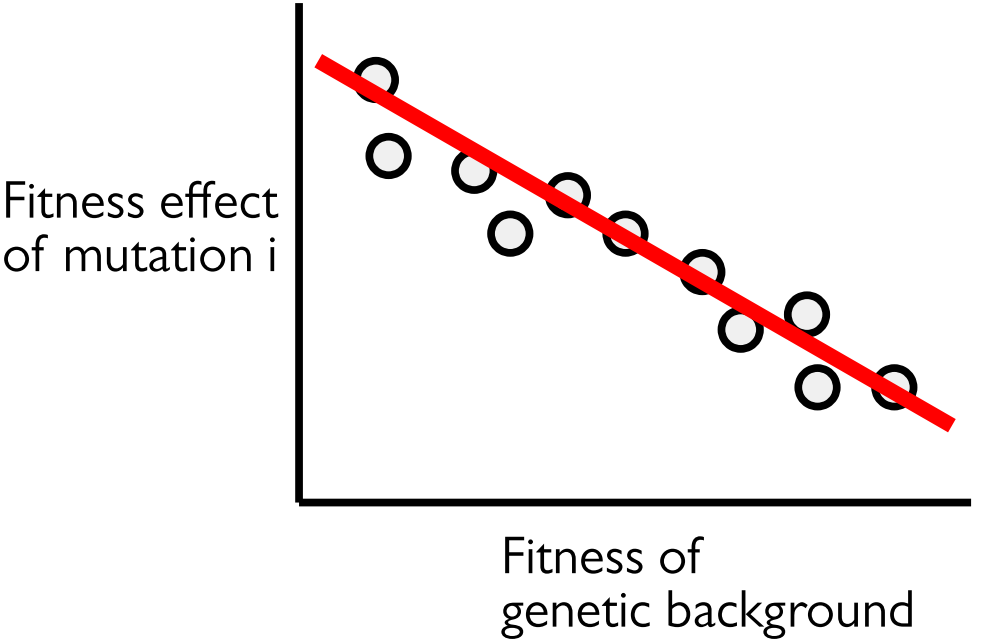




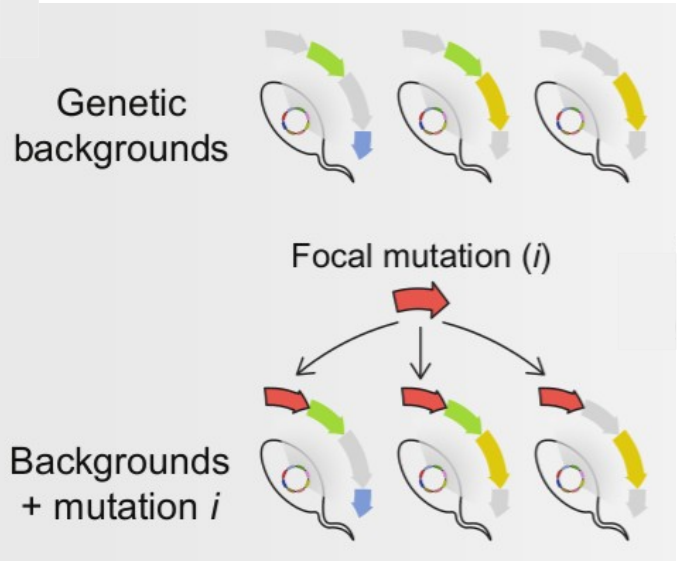
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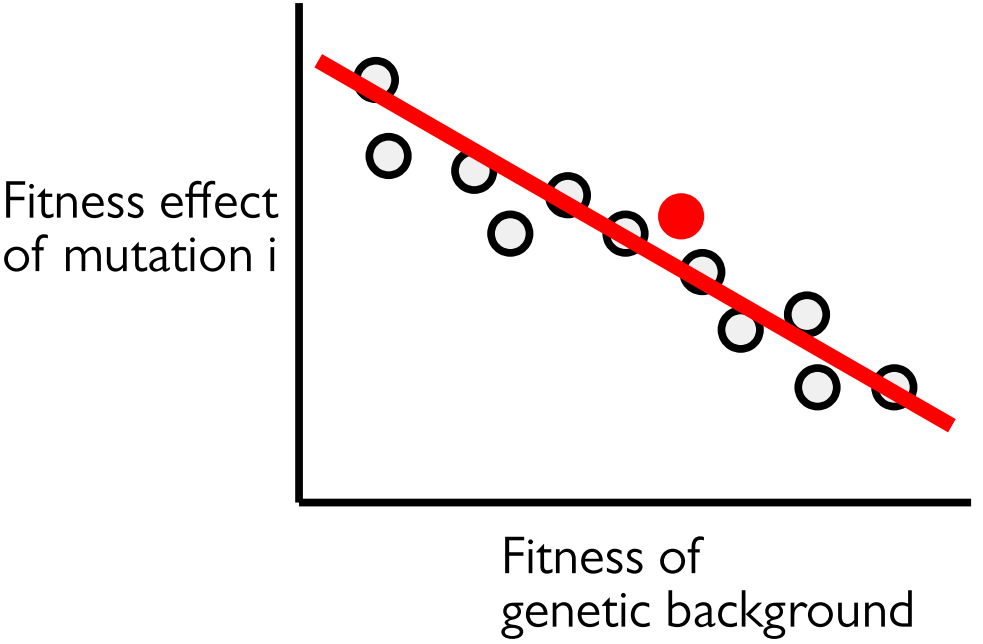
Global epistasis is an emergent linear scaling pattern found in many empirical fitness landscapes





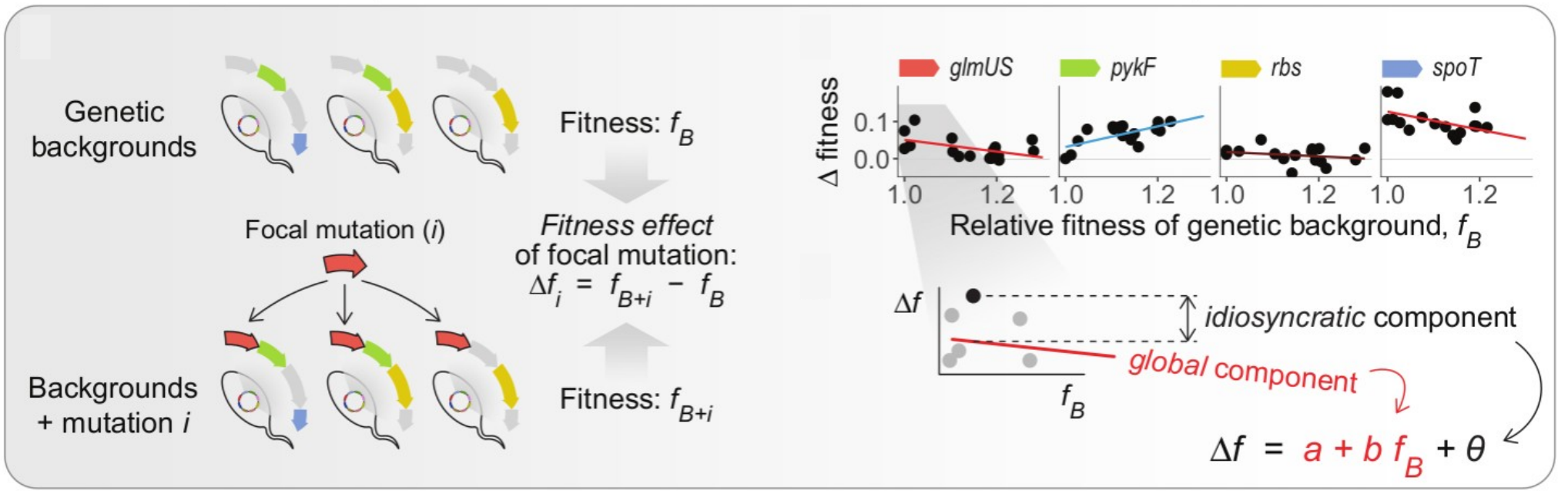
Global epistasis can help us make **predictions**





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Global epistasis is very common in empirical fitness landscapes

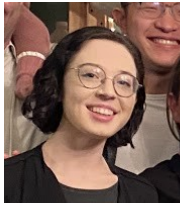


Does global epistasis exist also in structure function landscapes of microbial communities?

Does global epistasis exist also in structure function landscapes of microbial communities?



Juan Diaz Colunga (IPLA, CSIC)



Abby Skwara (Yale)

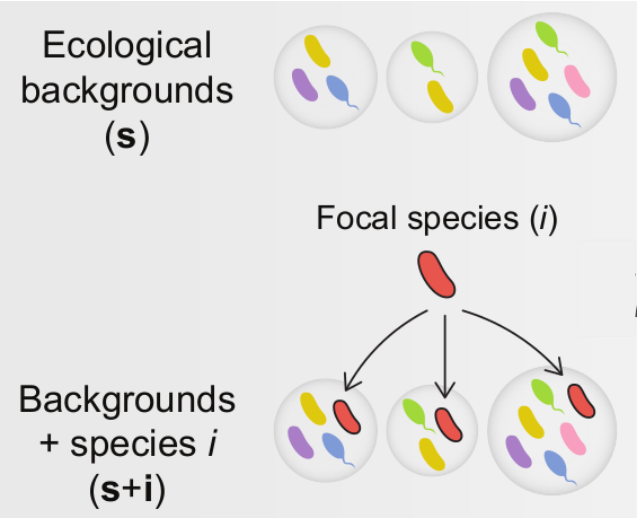


Álvaro Sánchez (CBGP, CSIC)



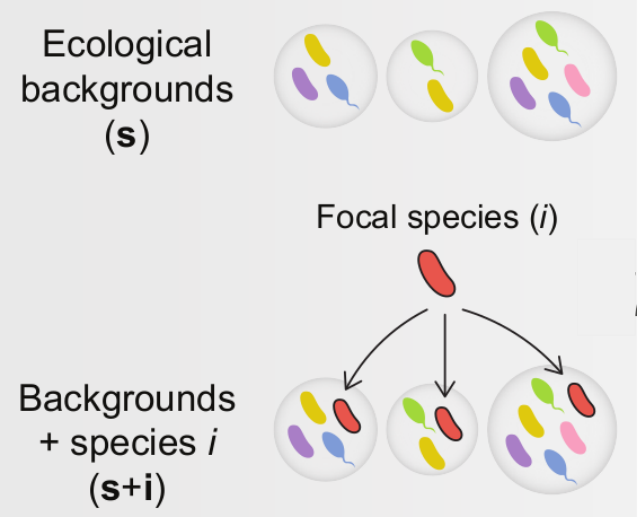
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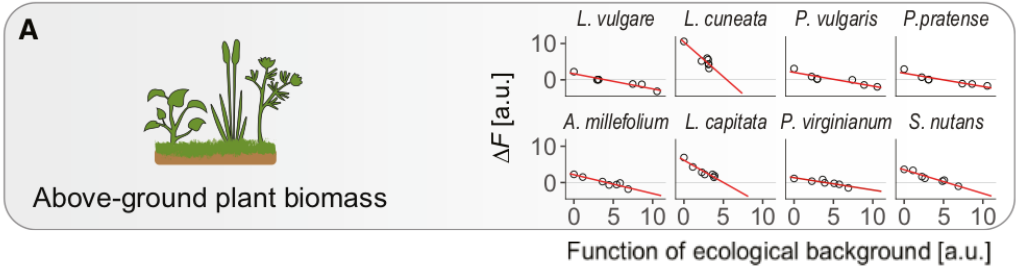
Is there global epistasis also in ecological structure-function landscapes?



Diaz-Colunga*, Skwara* et al, Cell, 2024

Is there global epistasis also in ecological structure-function landscapes?

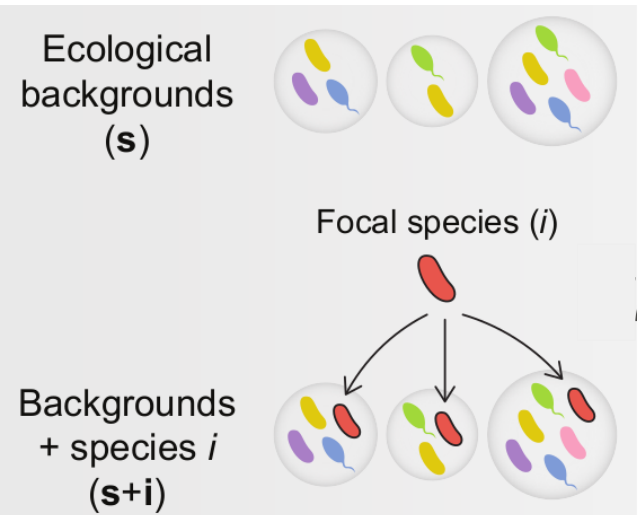


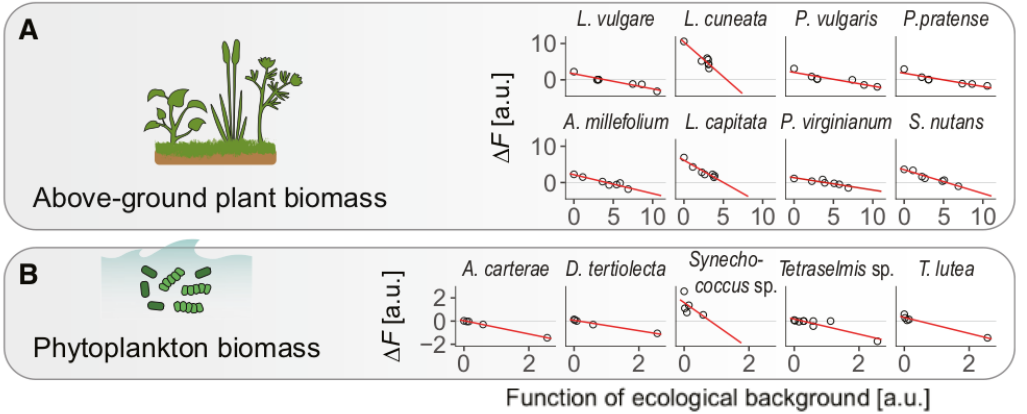


Diaz-Colunga*, Skwara* et al, Cell, 2024

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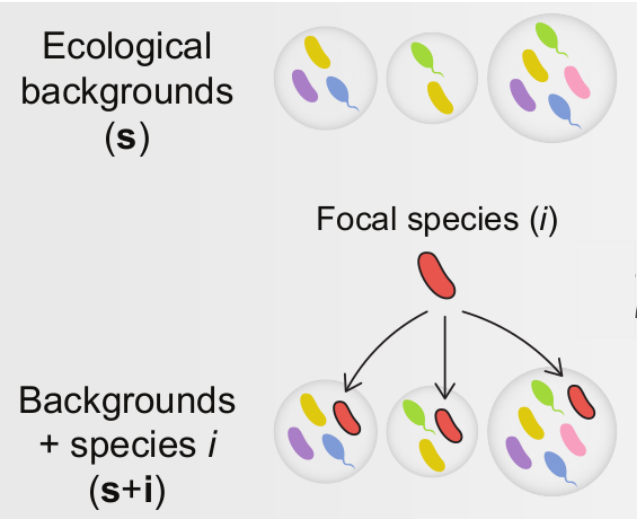


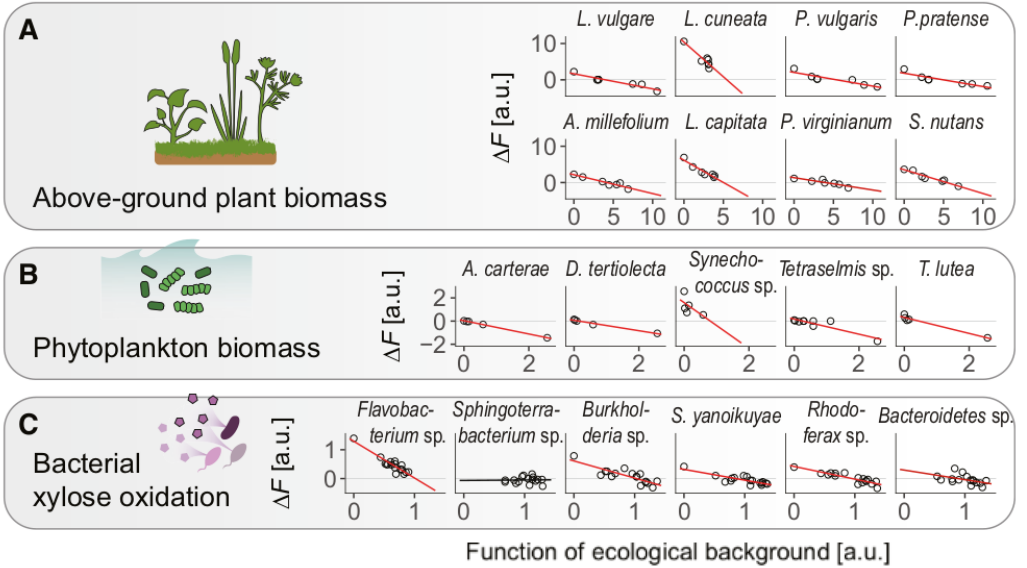


Diaz-Colunga*, Skwara* et al, Cell, 2024

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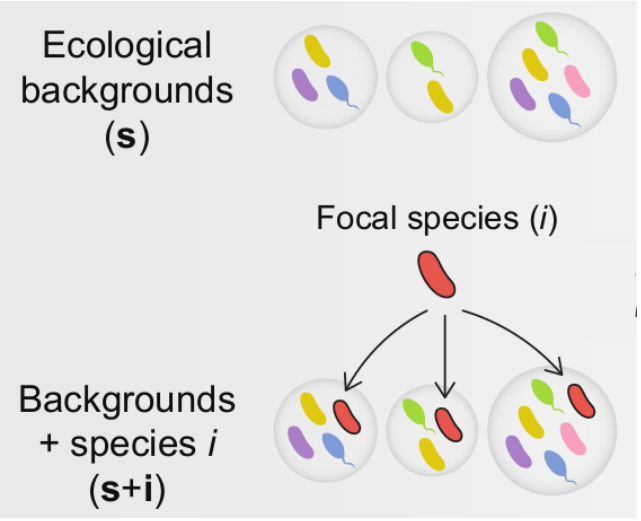


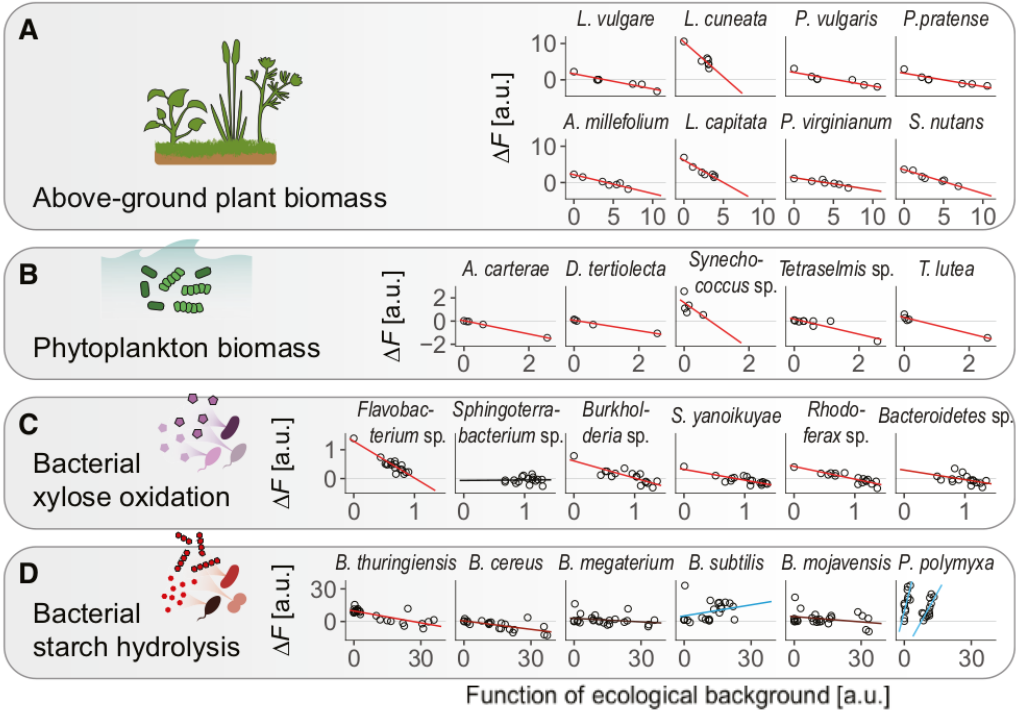


Diaz-Colunga*, Skwara* et al, Cell, 2024

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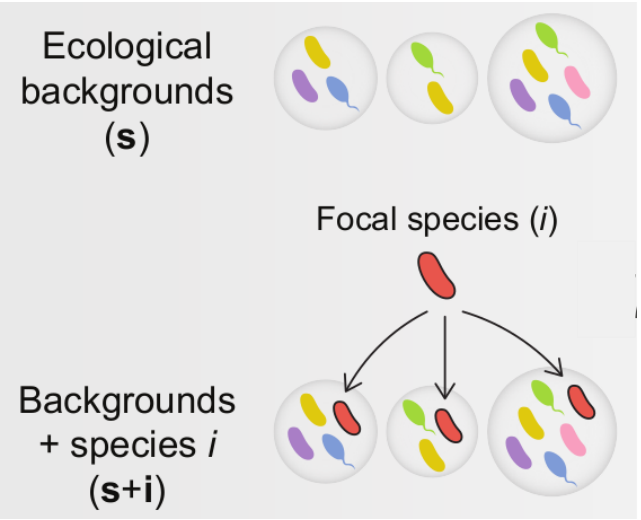


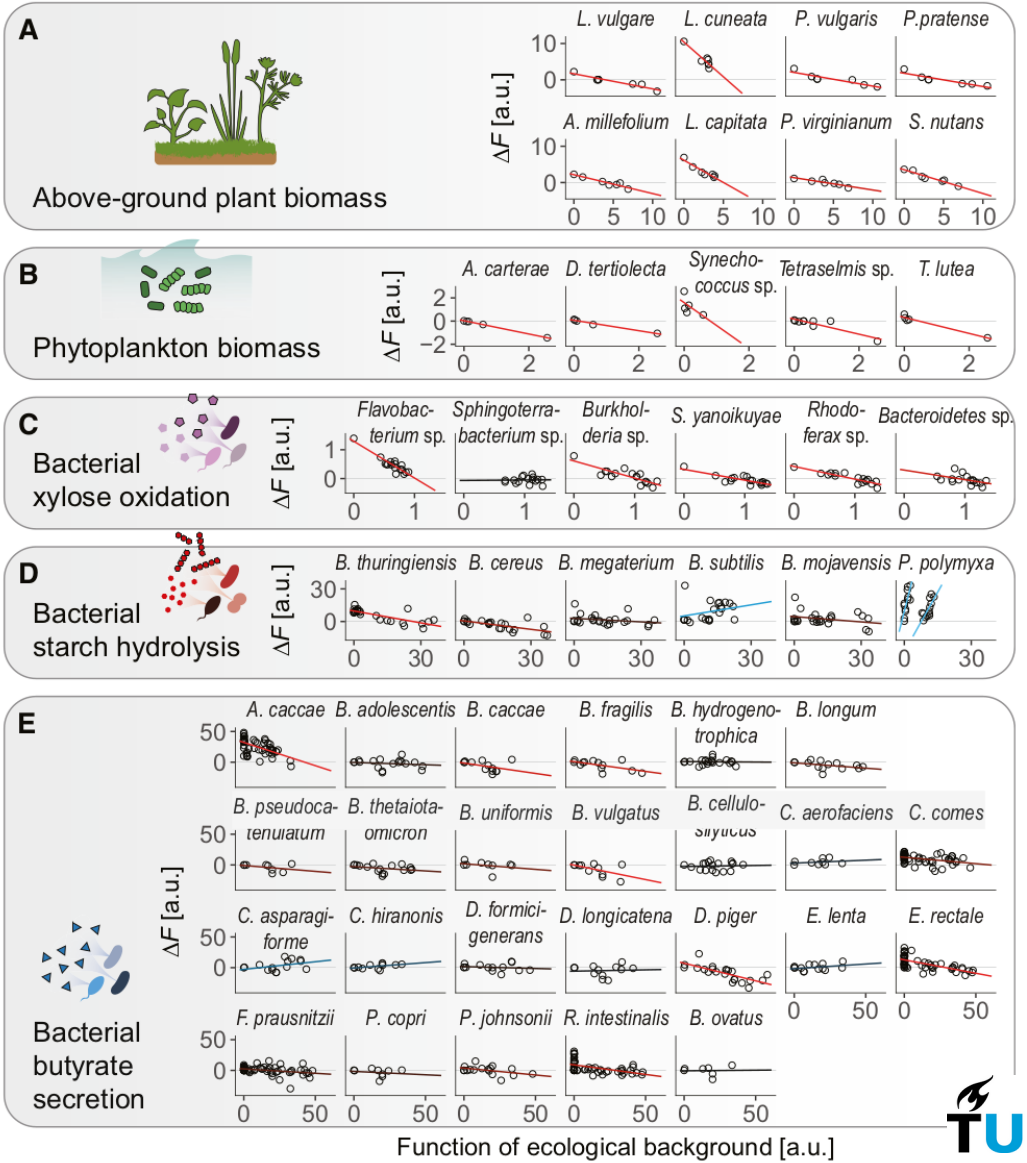


Diaz-Colunga*, Skwara* et al, Cell, 2024

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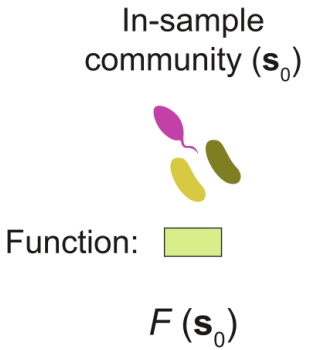




Diaz-Colunga*, Skwara* et al, Cell, 2024

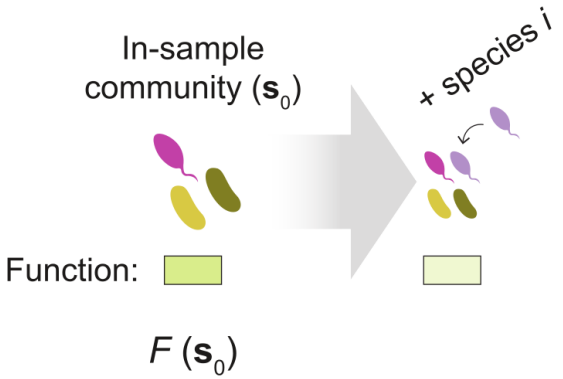
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How can we use global epistasis to make predictions?



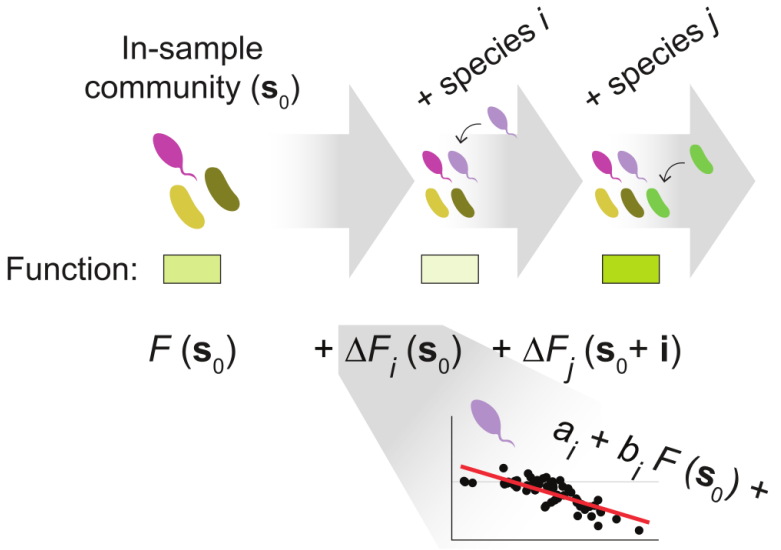
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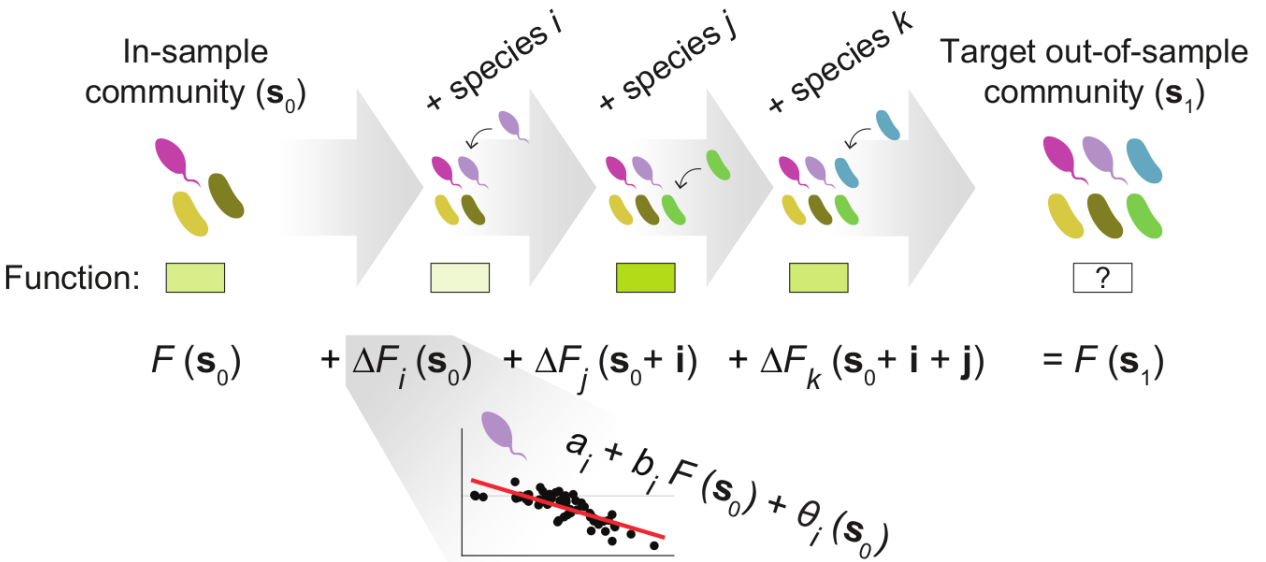
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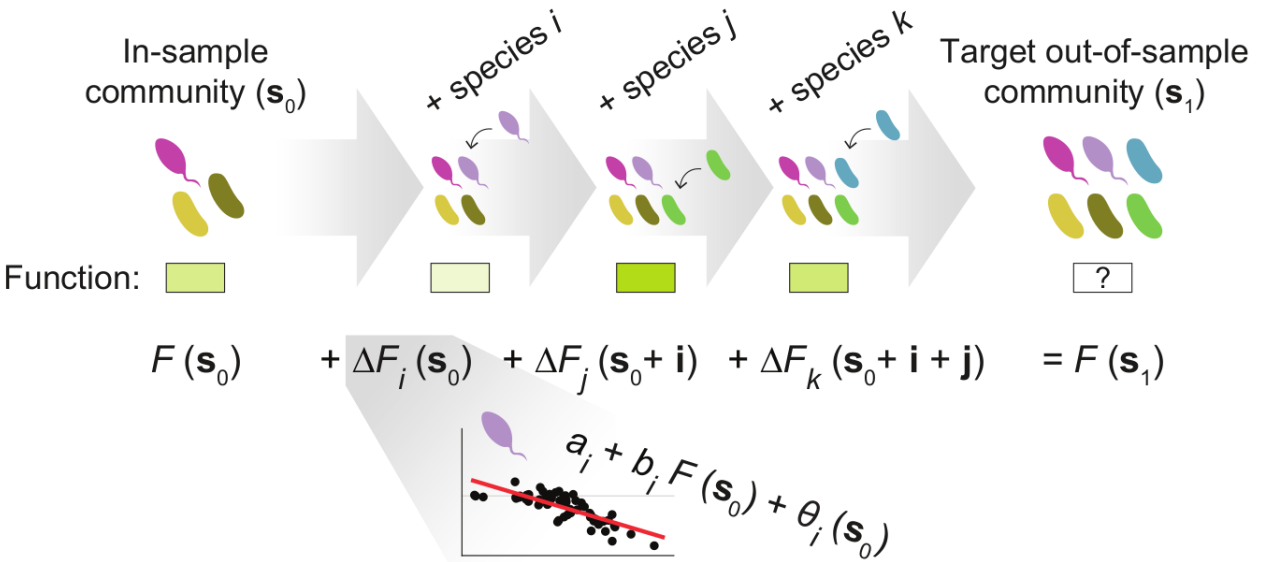
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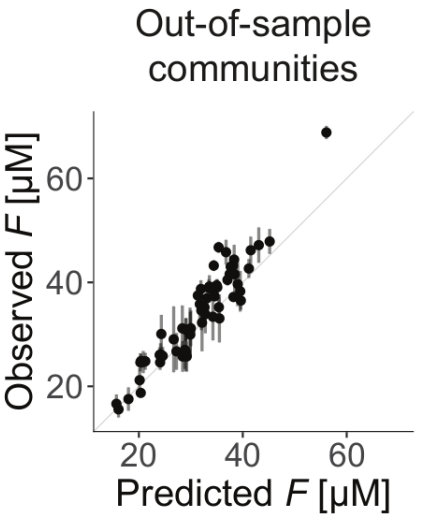


Diaz-Colunga*, Skwara* et al, Cell, 2024

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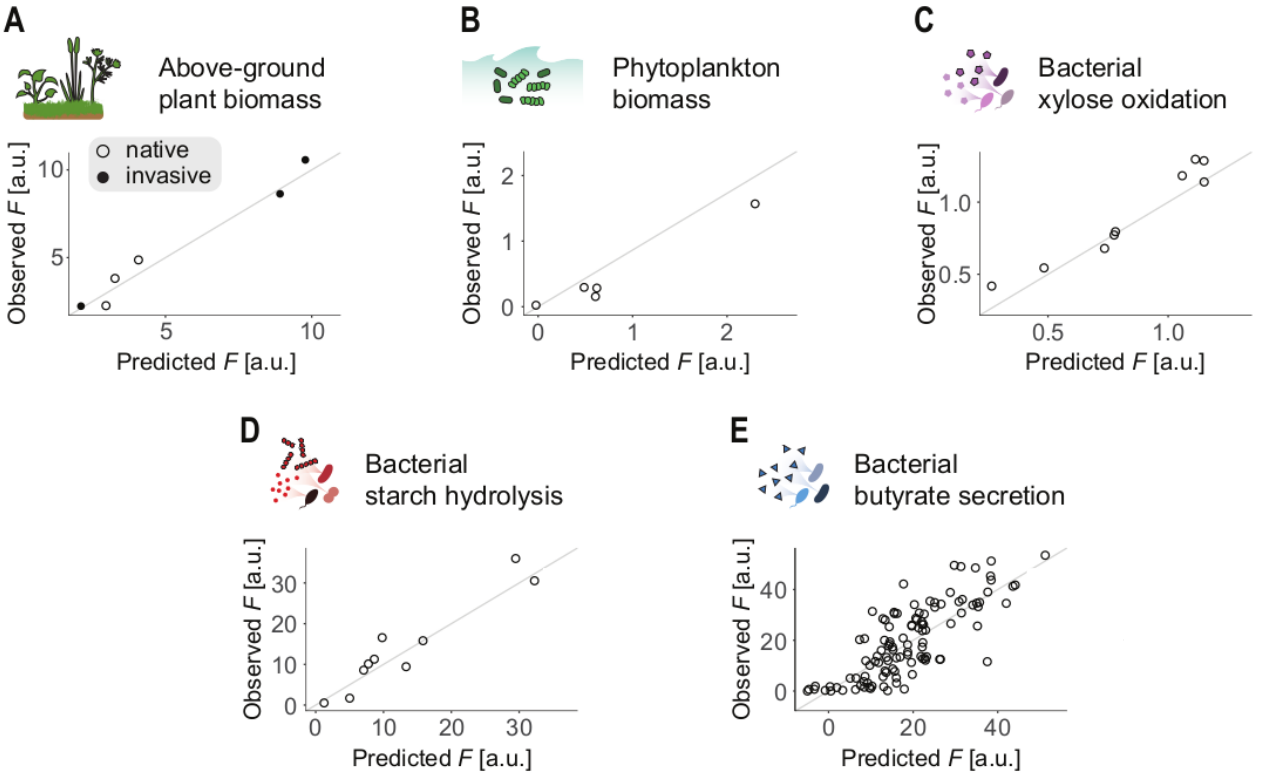


F = Pyoverdine production, R^2=0.81

Diaz-Colunga*, Skwara* et al, Cell, 2024

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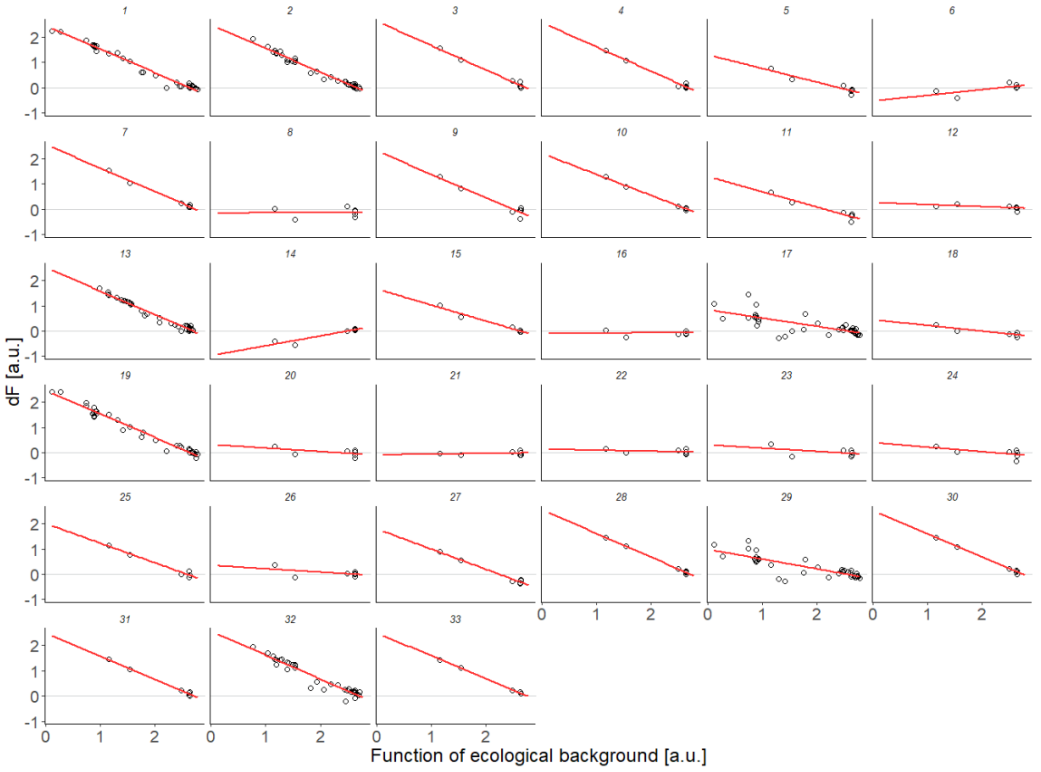
Global epistasis shows very promising predictive power

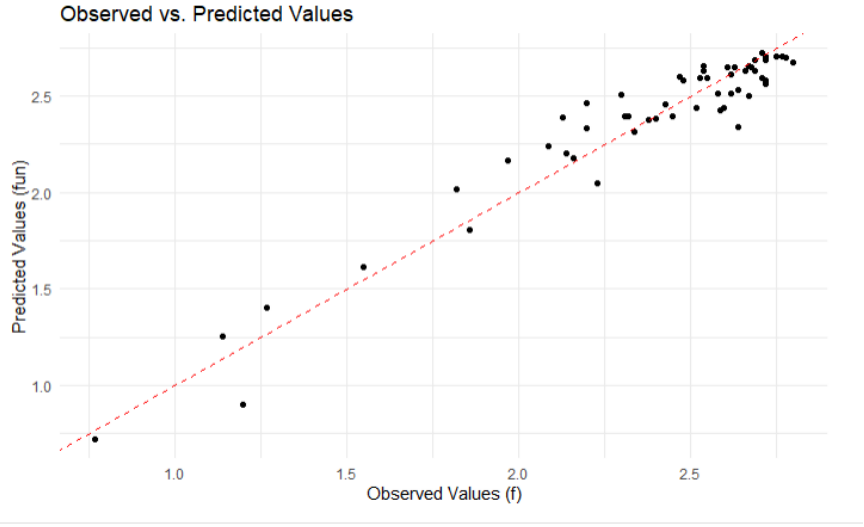


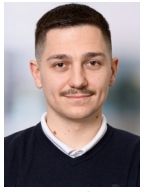
Diaz-Colunga*, Skwara* et al, Cell, 2024

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Global epistasis is able to predict **acidification** in synthetic soymilk-fermenting communities






Uros Gojković

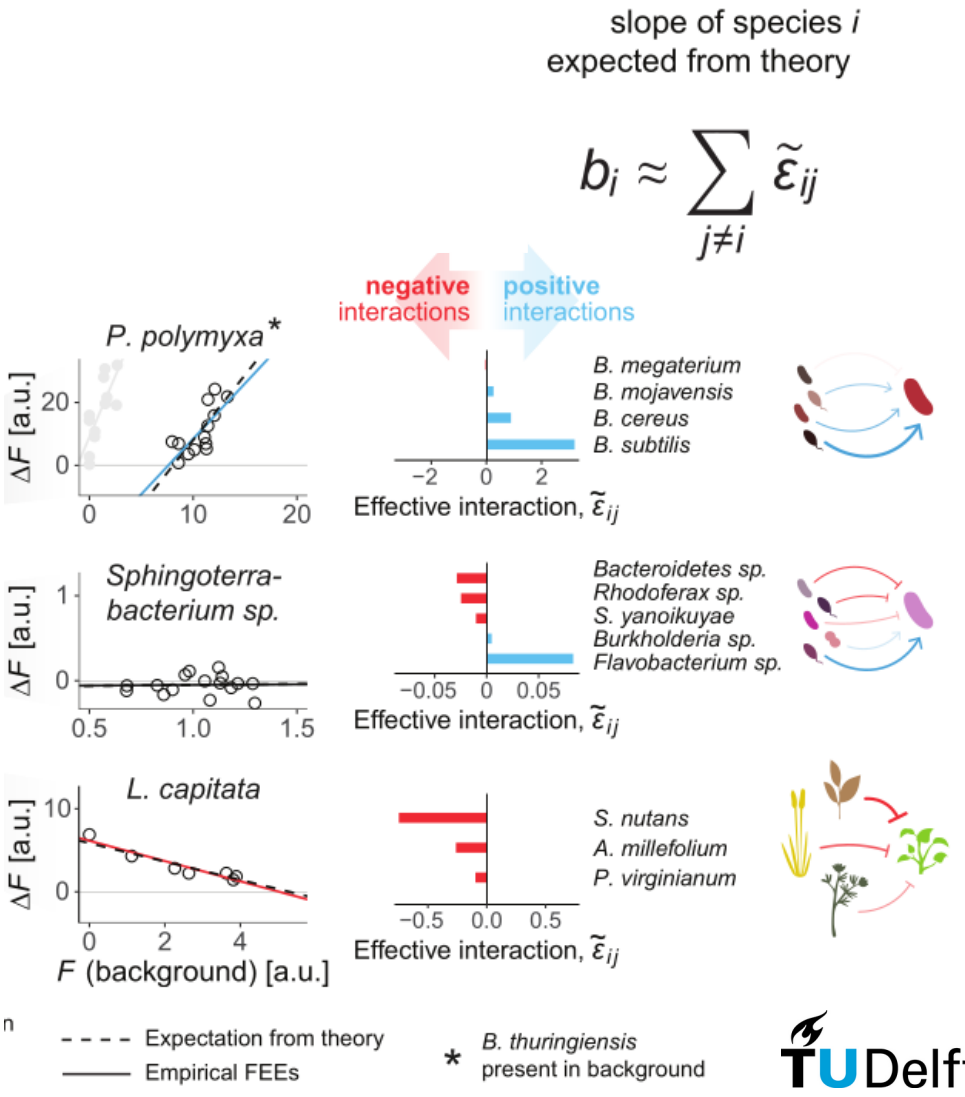

Nemanja Stanisavljevic (IMGGE)


Nemanja Kljajević (IMGGE)

Gojković et al, in prep

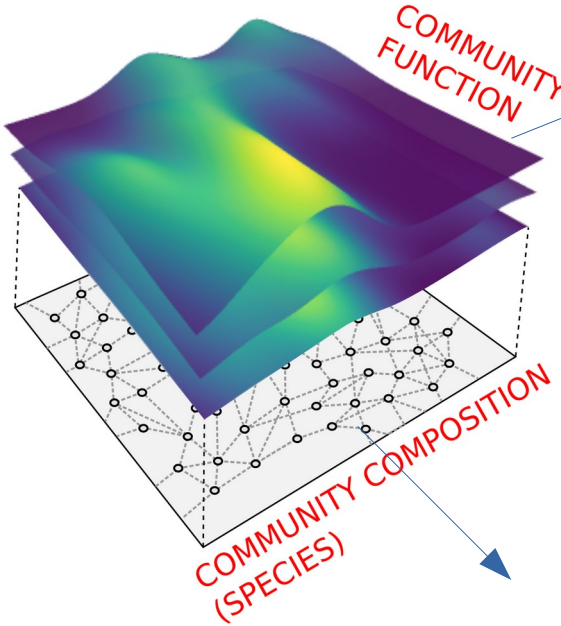
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Global epistasis is a spotlight to identify relevant interactions



Diaz-Colunga*, Skwara* et al, Cell, 2024

Can we generalize this even further,
beyond microbial communities?



Function: How much it kills *Staphylococcus aureus*?

Structure: composition of an antibiotic cocktail

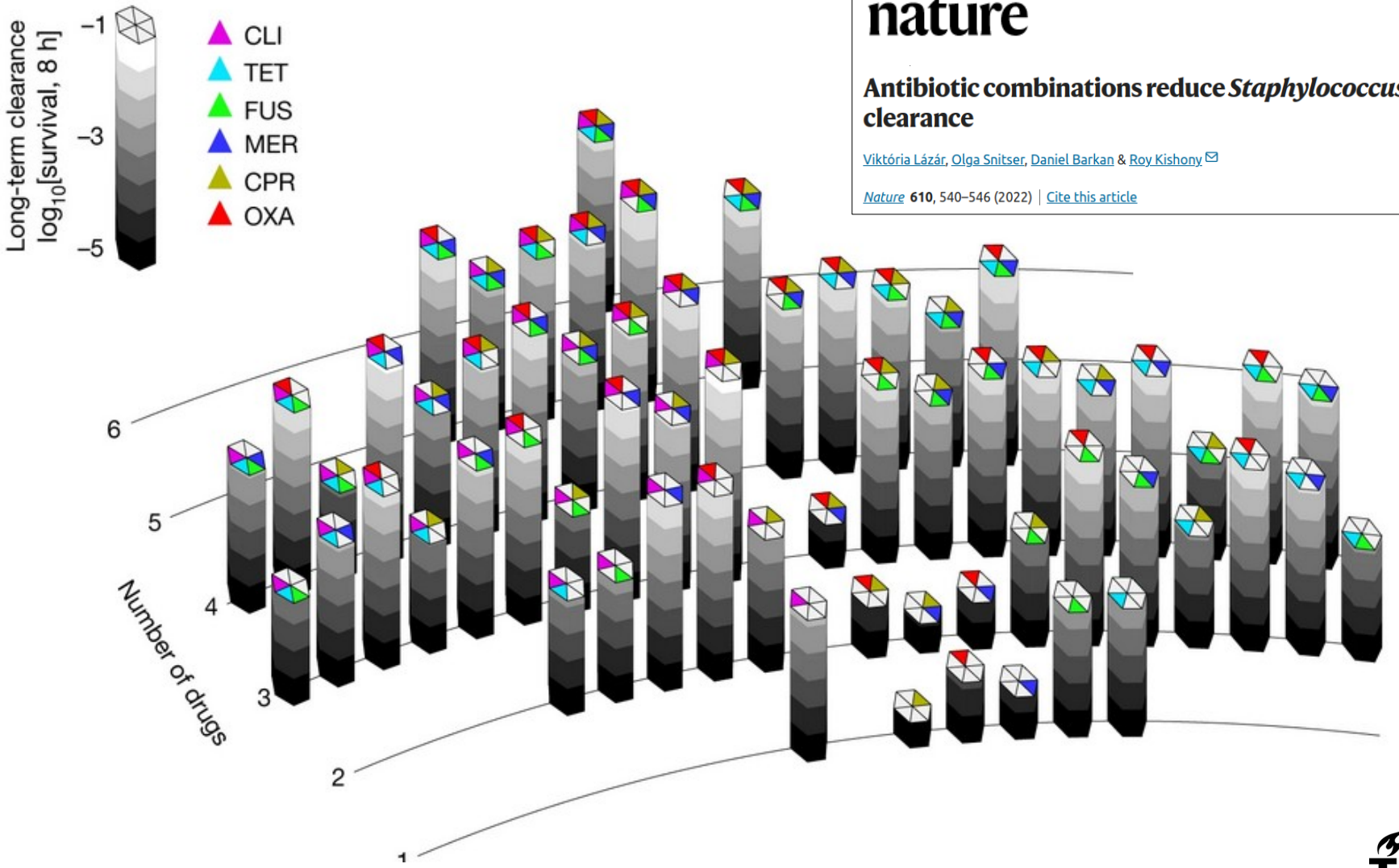


Floris Klein Obbink



Minke Gabriëls

TU Delft



nature

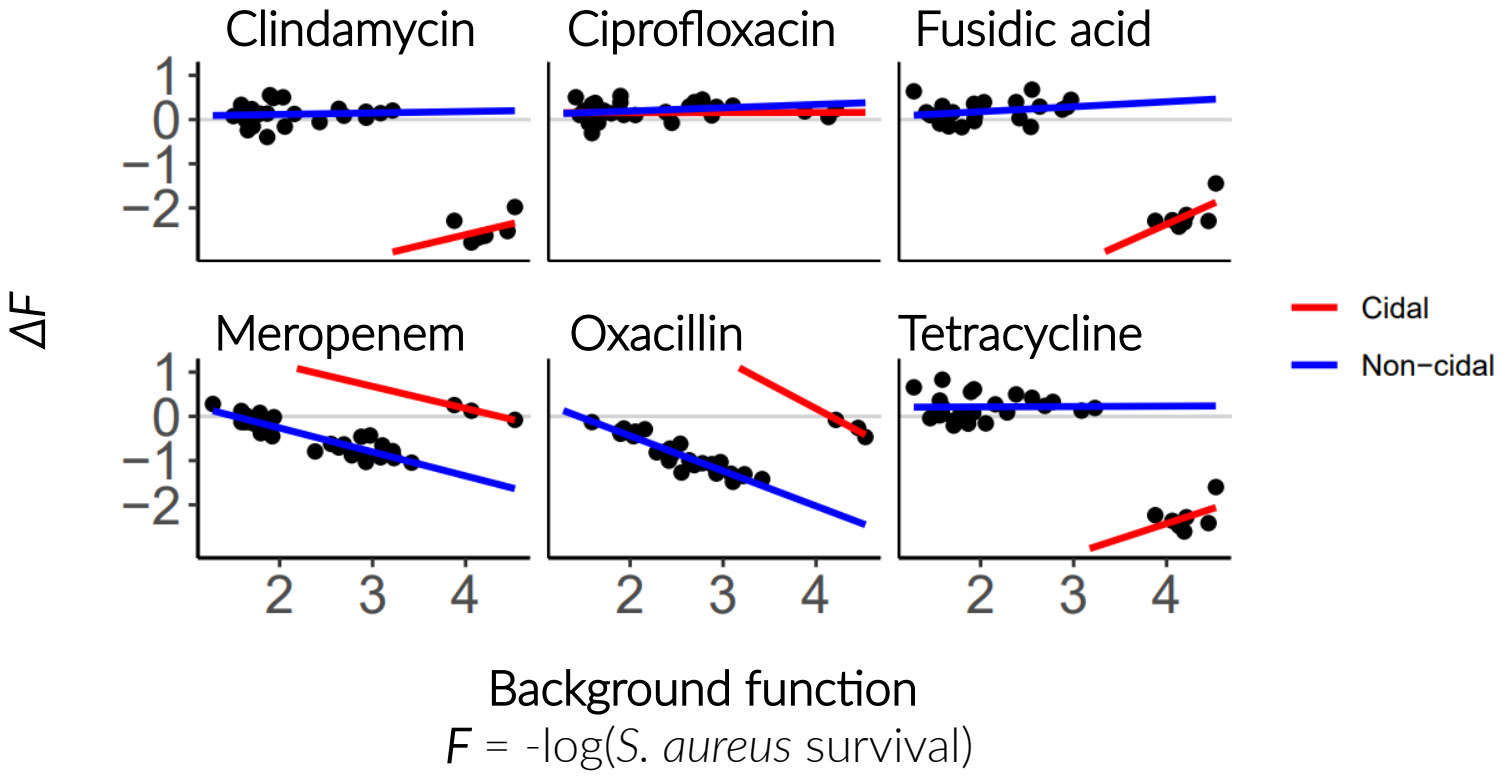
Antibiotic combinations reduce *Staphylococcus aureus* clearance

Viktória Lázár, Olga Snitser, Daniel Barkan & Roy Kishony ✉

Nature **610**, 540–546 (2022) | Cite this article

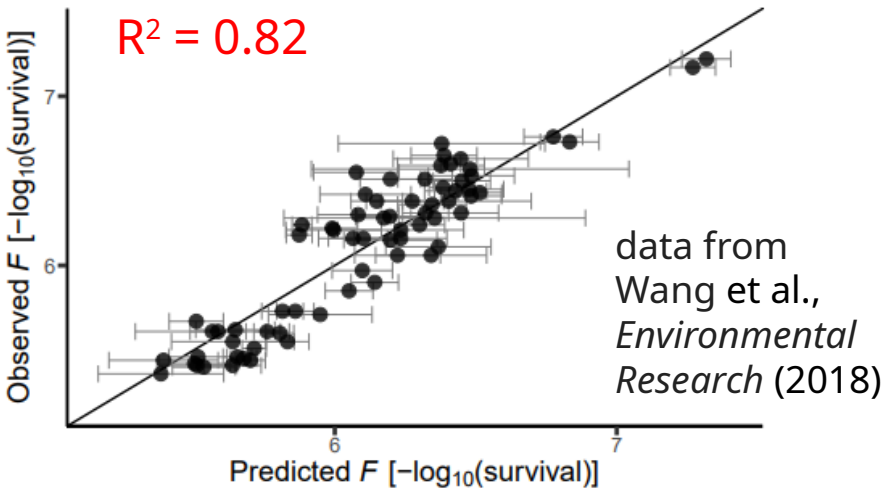
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Antibiotic combinations exhibit global epistasis in their effects on *Staphylococcus aureus* clearance

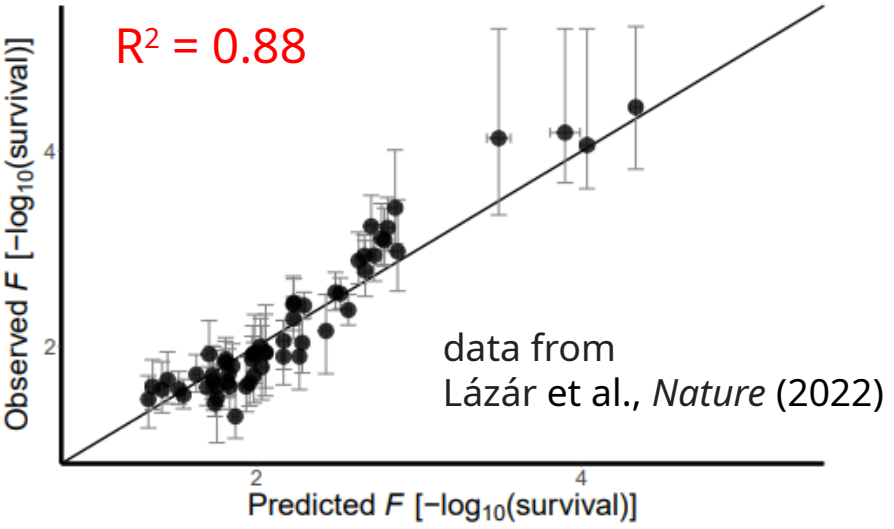


(atricle in prep)

The effect of unknown antibiotic cocktails can be predicted using global epistasis!



data from Wang et al., *Environmental Research* (2018)

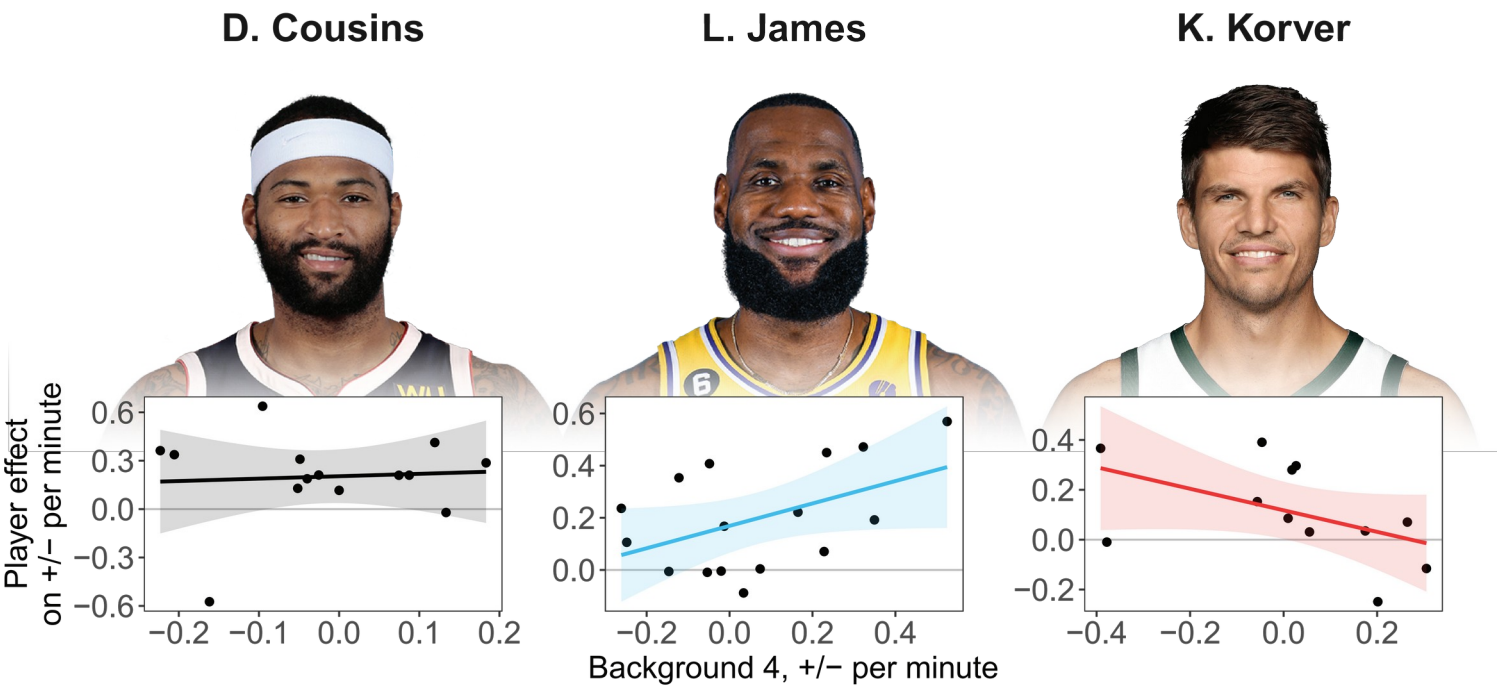


data from Lázár et al., *Nature* (2022)

(article in prep)

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And we can take it even further!



(Real data for NBA season 2015-16)

TU Delft

Collaborators

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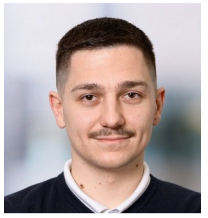
Funding

Zero Emission Biotechnology Program @ TU Delft

Thank you!

Group @ TU Delft


Minke Gabriëls


Uroš Gojković


Rienk Van Vliet

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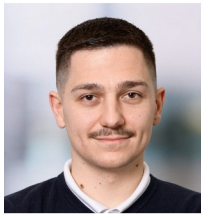
Funding

Zero Emission Biotechnology Program @ TU Delft

Thank you!

Group @ TU Delft


Minke Gabriëls


Uroš Gojković


Rienk Van Vliet

Practical Exercise

Statistical prediction and optimization of microbial community functions



Courtesy of: Juan Díaz-Colunga

Microbial Systems Biology Lab (MSBlab), Spanish National Research Council

10 minute break?